

Sex differences in bile acid homeostasis and excretion underlie the disparity in liver cancer incidence between males and females

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eLife Assessment

This study provides **valuable** insights into the influence of sex on bile acid metabolism and the risk of hepatocellular carcinoma (HCC). The data to support that there are inter-relationships between sex, bile acids, and HCC in mice are **convincing**, although this is a largely descriptive study. Future studies are needed to understand the interaction of sex hormones, bile acids, and chronic liver diseases and cancer at a mechanistic level. Also, there is not enough evidence to determine the clinical significance of the findings given the differences in bile acid composition between mice and men.

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Abstract

Hepatocellular carcinoma (HCC), the most common liver cancer, exhibits a higher incidence in males. Here, we report that mice lacking the bile acid regulators, Farnesoid X Receptor (FXR) and Small Heterodimer Partner (SHP), recapitulate the sex difference in liver cancer risk. Since few therapeutic options are available, we focused on understanding the intrinsic protection afforded to female livers. Transcriptomic analysis in control and FXR and SHP double knockout livers identified female-specific changes in metabolism, including amino acids, lipids, and steroids. We examined if the obtained transcriptomic signatures correlate with the survival outcomes for HCC patients to assess the translational potential of this

murine HCC model. Gene signatures that are unique to the knockout females correspond with low-grade tumors and better survival. Ovariectomy blunts the metabolic changes in female livers and promotes tumorigenesis that, intriguingly, coincides with increases in serum bile acid (BA) levels. Despite similar genetics, we found higher serum BA concentrations in males, whereas female knockout mice excreted more BAs. Decreasing enterohepatic BA recirculation using cholestyramine, an FDA-approved resin, dramatically reduced the liver cancer burden in male mice. Overall, we reveal that sex-specific BA metabolism leading to lower circulating BA concentration protects female livers from developing cancer. Thus, targeting BA excretion may be a promising therapeutic strategy against HCC.

Significance

We show that female-specific gene profiles identified in *Fxr*^{-/-}, *Shp*^{-/-} double knockout (DKO) mice correlate with better outcomes for HCC patients and uncover sex differences in circulation and excretion of bile acid. Overall, we demonstrate that increasing (ovariectomy or chemical injury) or decreasing (pharmacologically with FDA-approved resin) serum bile acids, not hepatic bile acids, promoted or alleviated liver cancer burden.

Introduction

Liver cancer, a leading cause of cancer-related death, has diverse etiologies and displays sex-difference with reduced risk in females compared to males [1–5]. Since current therapies for liver cancer fall short, we posit that understanding molecular mechanisms functioning in the female livers will reveal new therapeutic targets. Earlier studies have reported the role of sex hormones [6–9], transcription factors FoxA1/A2 [10], and cytokine IL6 signaling [11] in regulating the sex difference in hepatocellular carcinoma (HCC), but the role of metabolic pathways remains poorly understood.

Rewiring of cellular metabolism enables the tumor cells to maintain viability and grow disproportionately [12]. We previously showed that the combined deletion of nuclear receptors, Farnesoid X Receptor (FXR), and Small Heterodimer Partner (SHP) resulted in spontaneous liver cancer in the year-old male mice [13]. In this study, we report that, unlike the males, female *Fxr*^{-/-}, *Shp*^{-/-} double knockout (DKO) mice exhibit protection against tumorigenesis and thus mimic the sexual dimorphism in liver cancer incidence observed in clinics. Although 15-month-old individual *Fxr* knockout and individual *Shp* knockout mice were previously shown to develop liver cancer, unlike the DKO mice, their incidence does not show 100 percent penetrance nor sex differences [14–16].

Mutations and reduction in *Fxr*, and *Shp* transcript levels have been noted in cholestasis (reduced bile flow and subsequent increase in hepatic and serum bile acids (BA)), fatty liver disease, and liver cancer [17–23]. Moreover, individuals with chronic cholestasis exhibit an increased risk for HCC [24–26]. Typically, BA levels are tightly controlled via receptor signaling, including FXR and SHP [27–31]. Consistently, combined loss of *Fxr* and *Shp* in mice results in juvenile onset cholestasis that progresses to HCC [32]. We and others have shown that excessive accumulation and dysregulation of BA homeostasis are directly linked with liver cancer risk [13, 26, 33–35]. However, whether BAs are contributing factors to the sex differences seen in HCC prevalence has not been evaluated.

Therefore, we performed transcriptomic analysis to identify distinct gene profiles from both sexes of control and DKO mice. Then, using five separate human clinical HCC cohorts, we tested the clinical utility of the identified gene signatures from our mouse model. Next, we investigated the

role of endogenous estrogen signaling in the DKO mice by performing ovariectomy. We measured hepatic, serum, urine, and fecal BAs from male and female mice to understand their homeostasis. Finally, we manipulated the circulating BA levels in the DKO mice either with a chemical challenge or BA binding resins and examined its consequence on hepatic tumorigenesis. Overall, our data uncover that the differential BA homeostasis between the two sexes can orchestrate the observed gender differences in HCC burden in clinics.

Results

***Fxr*^{-/-}, *Shp*^{-/-} double knockout (DKO) mice phenocopy clinical features of HCC**

Here, we report that DKO mice exhibit the sexually dimorphic incidence of HCC observed in the clinic. Despite the loss of BA homeostatic machinery, one-year-old DKO female mice did not develop liver tumors but showed modest fat accumulation and mild fibrosis. On the contrary, DKO male livers revealed HCC and well-defined adenomas, robust steatosis, and fibrosis (**Fig. 1A-F**). Even at six months of age, female livers were smaller and displayed reduced hepatic fibrosis compared to males (**Supplementary Fig. 1**). Difference in tumor burden was reflected in the gross liver to body weight ratio, which was significantly higher in DKO males than DKO females (**Fig. 1G**). Serum ALT and AST were elevated in the DKO animals compared to WT (**Fig. 1H**) and correlated with the cholestatic phenotype of the mice. More importantly, liver cancer patients showed a reduction in *Fxr* and *Shp* transcript expression (**Fig. 1I**).

Sex-specific metabolic programs regulate liver tumorigenesis

To identify transcriptional mechanisms that can contribute towards the sex differences in the incidence of hepatic tumorigenesis, we analyzed one-year-old male and female livers. DKO males and females displayed striking differences in hepatic gene expression profile (**Fig. 2**, and GEO GSE151524), with DKO males showing enrichment of endoplasmic reticulum stress, unfolded protein response and immune function (**Fig. 2A-B**). Additionally, network analysis with ClueGO [36] revealed interactions between drug metabolism, inflammation, ERK signaling, and steroid metabolism in DKO males (**Fig. 2C**). On the contrary, DKO females displayed pathway enrichment of steroid metabolism and clustering of lipid, glucose, amino acid, and steroid metabolism, along with increased sulfotransferase activity (**Fig. 2D**). Next, we parsed the sex-specific upregulated gene sets to identify unique transcription factor motifs. Overrepresented motifs in DKO males, including AR, FOXA1, FOXA2, NRF2, and PPAR γ [10, 37–40], correlated with tumor-promoting functions (**Supplementary Table 2**). In contrast, in DKO females, FOXO1, E2F, and ER α (**Supplementary Table 3**) were dominant motifs and are associated with regulating metabolic function during liver carcinogenesis [41].

The transcriptomic signature of the DKO mice correlates with poor overall survival in the clinical datasets

To investigate the clinical relevance of the DKO mouse model, we analyzed the WT and DKO murine transcriptomic signatures in a sex-specific manner and compared these to five separate clinical HCC datasets (**Supplementary Table 4**). The patient data were sorted based on similarity to one-year-old DKO gene signatures using class prediction (**Supplementary Fig. 2–3**). Computational prediction scores (BCCP: 1 represents complete match and 0 represents no match) using the patient samples revealed that the DKO male signature matched with the later stages (≥ 2) of liver cancer, whereas the DKO female-specific signature matched well with earlier tumor stages (CLIP score 0 or TNM stage 1) (**Supplementary Fig. 3**).

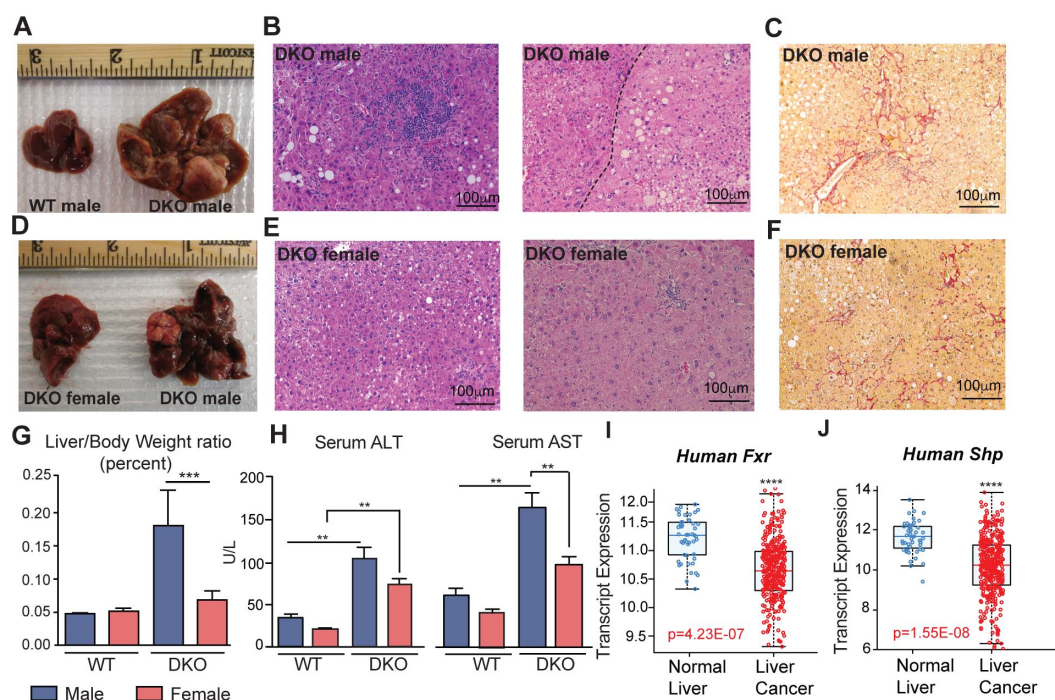


Figure 1

***Fxr/Shp* double knockout (DKO) mouse model recapitulates sex difference observed in HCC incidence.**

(A-E) One-year-old DKO male mice developed hepatocellular carcinoma, which was not observed in age-matched wild type (WT) and DKO female mice. (B and E) Representative H&E stained liver sections from a (B) DKO male and (E) female. Inflammation and injury are evident at 1 year, and the dotted line (B) separates the HCC with large nuclei on lower right. Sirius red staining shows increased collagen in a perisinusoidal distribution, which is greater in the DKO males (C and F). The liver-to-body weight ratio was significantly higher in DKO males (G). Compared to WT animals, serum markers of liver injury, (H) AST and ALT were higher in DKO mice. (I-J) Analysis of five different HCC clinical cohorts (n=1000) patients reveals a reduction in *Fxr* and *Shp* transcript levels in patients with liver tumors. n=5-10 mice /group; mean $\pm$ SEM; * $p<0.01$, ** $p<0.001$ compared to genotype or gender controls. One-way ANOVA with Bonferroni post hoc analysis was performed.

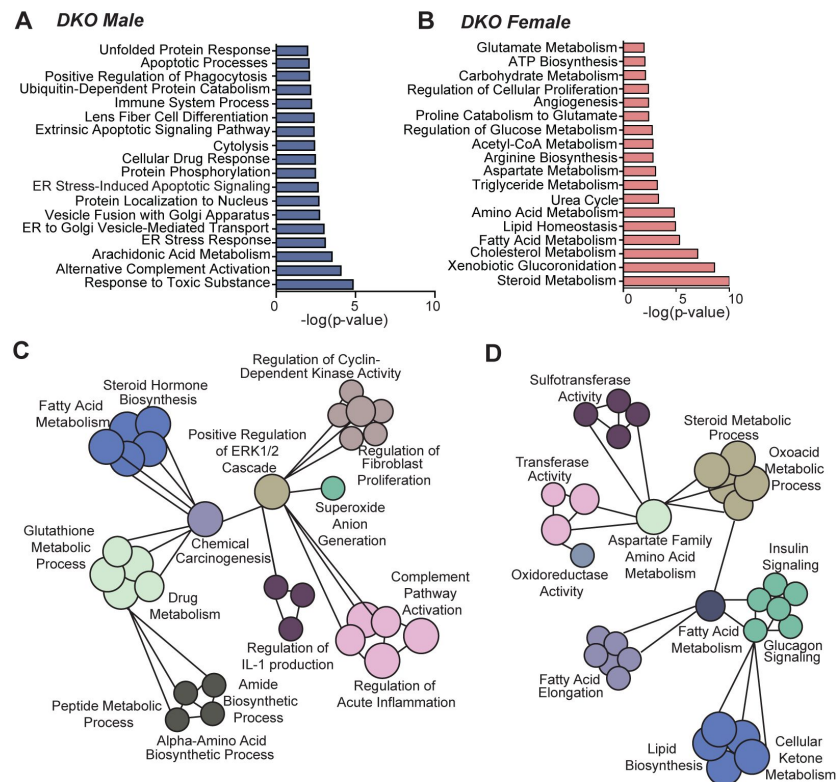


Figure 2

Transcriptome analysis reveals striking sex differences in hepatic metabolism.

Microarray was performed on liver tissue from WT and DKO mice of both sexes ($n = 6/\text{group}$). (A-B) GO categories were determined using genes with >1.3 fold change in expression between DKO males and females. Enrichment of overlapping GO categories between males and females was determined by comparing $-\log p$ -values for each term. (C) GO categories unique to the set of genes upregulated >1.3 fold in DKO males and (D) DKO females.

Of the 1100 patient data, we found approximately (~45%) showed a transcriptomic signature similar to that of either DKO male or DKO female, which corresponded to lower overall survival (OS), but not recurrence-free survival (RFS). WT gene signatures were used as controls (**Fig. 3A-B**). Although DKO female mice do not develop liver cancer, it is pertinent to note that they lack *Fxr* and *Shp* expression, display chronic cholestasis similar to their male counterparts, and hence the global gene changes associate with poor OS. On the contrary, when we focused on the gene signature that was distinctly changed only in the DKO female livers, not the DKO males, we found that patients (~54.71%) who displayed this subset of gene signature had better OS as well as RFS (**Fig. 3C**). These findings underscore a high potential for clinical translation of data generated from the DKO mouse model. Moreover, by focusing on specific transcript changes in the DKO female livers, we uncovered a subset of metabolic genes that correspond to better survival and might be responsible for their protection against cancer.

We initially examined the pathways pertaining to amino acid metabolism and ureagenesis since individuals with mutations in the urea cycle disorder have an increased risk of developing liver cancer [42–45]. Consistent with this, analysis of the TCGA-LIHC clinical dataset revealed a broad downregulation of genes encoding the entire urea cycle, including carbamoyl phosphate synthetase (*Cps1*), ornithine transcarbamylase (*Otc*), argininosuccinate synthetase (*Ass1*), argininosuccinate lyase (*Asl*), and arginase (*Arg1*) in both sexes upon liver tumorigenesis (**Supplementary Figure 4**). In contrast, these genes were all upregulated in DKO female livers (**Supplementary Fig. 5A**), which correlated well with the protection afforded to the DKO female livers as loss-of-function mutations in these genes are linked to HCC [45–47]. Additionally, our analysis showed that patients with increased expression of urea cycle genes (DKO-UreaCycle) exhibited a better clinical outcome (**Supplementary Fig. 5B**).

Estrogen signaling controls amino acid and bile acid metabolism in the liver

Since estrogen signaling was previously shown to regulate amino acid metabolism [48], we examined its role in controlling the expression of urea cycle genes in the DKO female livers. To do this, we ovariectomized (OVX) DKO mice and found that, indeed the hepatic expression of all these genes, *Cps1*, *Asl1*, *Ass*, *Otc* and *Arg1* were significantly blunted in the absence of endogenous estrogen signal (**Supplementary Fig. 5C**). But when we measured the urea cycle metabolites, we did not find any significant change in the intermediate nor urea production except for a decrease in ornithine levels (**Supplementary Fig. 5D**), in DKO females compared to the DKO males. We reason that static measurements may not reflect the flux into the urea cycle.

Besides amino acid metabolism, estrogen signaling has been shown to affect BA homeostasis and cause cholestasis [49–51]. So, we anticipated that OVX would lower BA levels in DKO female mice. Instead, we found that OVX led to liver cancer development in otherwise resistant year old DKO female mice (**Fig. 4A-B**). Moreover, their serum BA levels doubled (**Fig. 4C**), consistent with the tumorigenic role of BAs. Also, analysis of TCGA-LIHC clinical data revealed significant downregulation of *Era* gene expression in liver tumors (**Supplementary Figure 6A**). In addition, estrogen signaling gene signature obtained from the DKO livers correlated with better overall- and recurrence-free survival (**Supplementary Figure 6B**). Importantly, these results corroborated well with clinical observations that post-menopausal women exhibit higher susceptibility to developing HCC, which can be mitigated upon hormone replacement therapy [52, 53].

To overcome the confounding effects of ageing and cancer, we examined young WT and DKO female mice with and without OVX. Additionally, we challenged these mice with BA excess (**Fig. 4D**). As expected, OVX resulted in the reduction of basal hepatic *Era* gene expression in both WT and DKO mice (**Fig. 4E**). In the DKO mice, which display high basal levels of BA synthesis and sulphation genes, we found dramatic induction of *Era* gene upon BA treatment (**Fig. 4E**). Importantly, the rise in *Era* gene coincided with reduced expression of *Cyp7a1*, *Cyp8b1* and

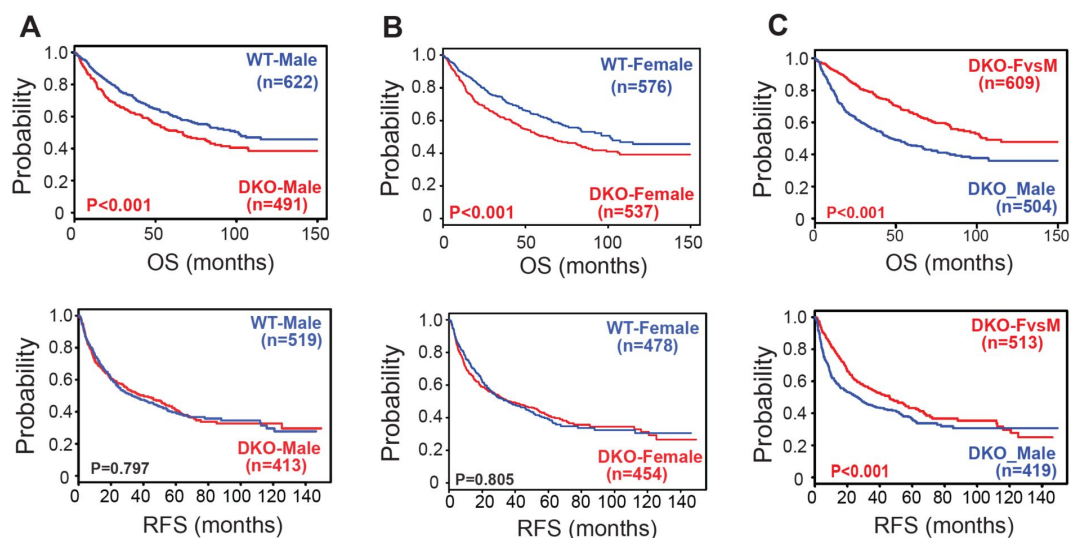


Figure 3

Gene signatures obtained from the DKO mouse model correlate well with the clinical outcomes of HCC patient.

The survival probability based on WT and DKO transcriptome changes was evaluated using five different HCC clinical cohorts. (A-C) Analysis of OS (Overall Survival) and RFS (Recurrence Free Survival) in patients using the gene signatures representative of either male WT or DKO mice (A), female WT or DKO mice (B), and unique changes observed in female DKO mice (C).

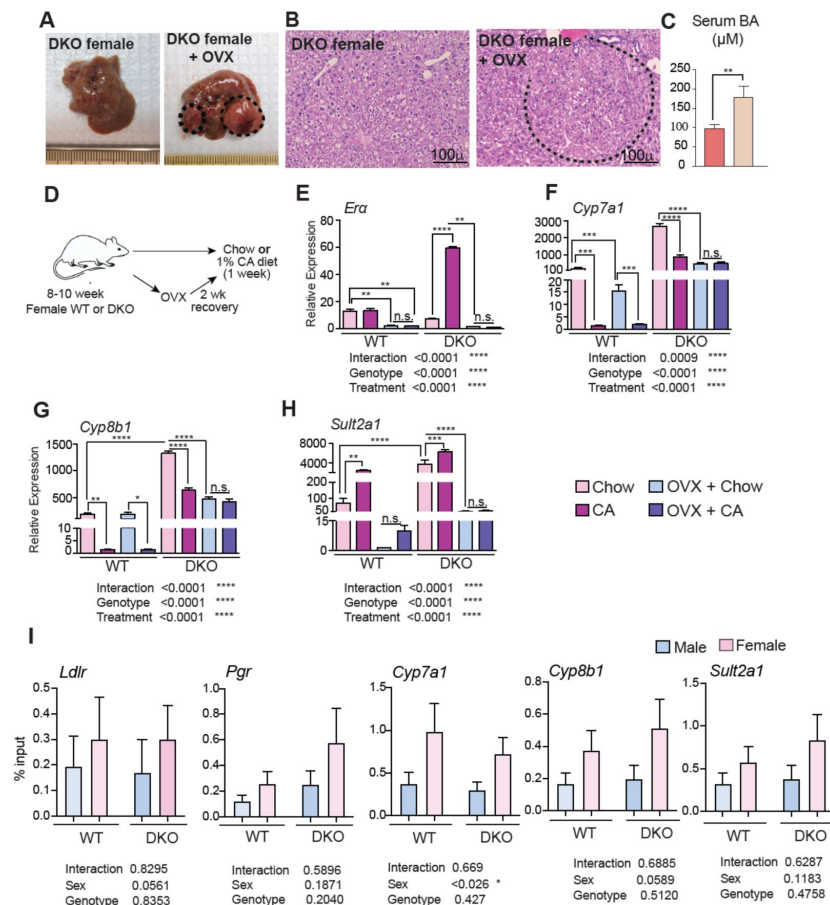


Figure 4

Estrogen signaling protects against liver tumorigenesis and can regulate BA synthesis in DKO female mice.

(A-B) Ovariectomized female DKO mice were aged to a year and examined for liver tumorigenesis, where a dotted line demarcates the tumor margin. (C) Serum total bile acid concentrations. (D) Experimental design of chow and 1% cholic acid (CA) diet for 1 week with or without (OVX). (E) Expression of hepatic *Era* was induced with CA diet in DKO female mice and reduced in both WT and DKO females following ovariectomy. (F) CA-mediated suppression of *Cyp7a1* and (G) *Cyp8b1* in WT and DKO females was lost in DKO females after OVX. (H) *Sult2a1* has greater baseline expression in DKO mice, induced to a lesser extent upon CA challenge compared to WT animals ($n = 4-5/\text{group}$). (I) ChIP-PCR was performed in WT and DKO male and female livers to test ER α recruitment to BA synthesis and metabolism genes, *Cyp7a1*, *Cyp8b1*, and *Sult2a1*. Mean $\pm$ SEM; Two-way ANOVA with Bonferroni post hoc analysis was performed. # $p < 0.05$, * $p < 0.01$, ** $p < 0.001$ compared to controls.

increased levels of *Sult2a1*, a sulphotransferase known to sulphate estrogen and BAs. OVX in WT mice led to lower basal levels of *Cyp7a1* and *Sult2a1* but not *Cyp8b1*, whereas all three genes were significantly reduced in the DKO livers (**Fig. 4G–4H**). Unlike the OVX WT, which maintained CA-mediated suppression of BA synthetic genes, consistent with intact FXR signaling, DKO+OVX mice did not alter their expression (**Fig. 4F–G**). These data indicate a role for estrogen signaling in regulating BA homeostasis in the DKO livers.

We next examined if the recruitment of ERα to BA synthesis genes exhibited any sex difference in WT and DKO livers by ChIP-PCR. We find that ERα was preferentially recruited to *Cyp7a1* in a sex-specific manner (**Figure 4I**). *Cyp8b1* showed a similar trend but not *Sult2a1*. Also, we did not find any sex-specific patterns in ERα occupancy in *Ldlr* and *Pgr* genes, which were used as positive controls for ERα ChIP assays (**Figure 4I**). These data, along with increased BAs upon OVX, suggest ERα signaling is pertinent to control BA synthesis, especially in the absence of FXR, as seen in the CA-fed sham DKO mice.

DKO mice display sexual dimorphism in BA homeostasis

WT mice do not show overt changes in serum BAs between the two sexes; however, genetically identical DKO mice displayed dramatically lower serum BAs in females compared to males (**Supplementary Table 5**). Nonetheless, the serum BA concentration in DKO females was higher than in WT females. This was intriguing. So, we analyzed the expression of genes involved in BA synthesis, transport, and metabolism in both sexes of DKO mice. Consistent with *Fxr* and *Shp* deletion that results in the loss of negative feedback on BA biosynthesis, both sexes of DKO mice have significantly higher expression of *Cyp7a1* and *Cyp8b1* genes that are involved in classical BA synthesis (**Fig. 5A**). The male dominant expression of *Cyp7b1* in the WT is lost in the DKO mice. On the other hand, *Cyp27a1*, which initiates alternative BA synthesis, was increased in a female-specific manner (**Fig. 5A**).

Next, we examined BA transport. We found that hepatic transcript levels of the key BA efflux pump, bile salt export pump (*Bsep*, *Abcb11*), were reduced in both sexes of DKO mice, consistent with loss of *Fxr* (24) (**Fig. 5B**). In contrast, the expression of canalicular efflux transporters, *Abcb1* (*Mdr1*) and *Abcc2* (*Mrp2*) was unchanged (**Fig. 5B**). Also, the BA uptake transporter, sodium taurocholate co-transporting polypeptide (*Ntcp*, *Slc10a1*) showed lower transcript levels in females (**Fig. 5B**), which is in line with previous findings that estradiol can downregulate *Slc10a1* expression [54].

We then investigated the transcript expression of *Sult2a1*, which contributes to BA sulfation—a modification that can reduce enterohepatic recirculation [55]. As expected, hepatic *Sult2a1* expression was predominant in females irrespective of the genotype (**Fig. 5C**) [56]. Sulphated BAs are excreted in urine to eliminate excess BA during cholestasis [57, 58]. Total urine BA levels were higher in DKO males, reflecting a larger circulating BA pool than in DKO females (**Supplementary Table 6**). However, DKO female mice exhibited higher percentages of sulphated BAs (**Fig. 5D**), which corroborates with high *Sult2a1* expression in females.

BA compositional analysis was performed in the serum, hepatic, urine, and feces of DKO males and females (**Fig. 5E–H**). Both sexes of DKO mice showed abundant muricholates in the serum (**Fig. 5E**), but there were modest differences in the composition, indicating slightly hydrophilic BAs in the DKO females. Moreover, hepatic BA composition was indifferent between the two sexes (**Fig. 5F**). These results indicated that rather than synthesis or transport, excretion may be different between DKO males and females. Notably, we found that both urine and fecal levels and composition between male and female DKO mice were distinct (**Fig. 5G–H**). As urinary BA excretion alone cannot explain the 50% decrease in circulating BAs in DKO females, we performed untargeted metabolomics using the fecal samples. BAs accounted for 20% of the fecal samples in

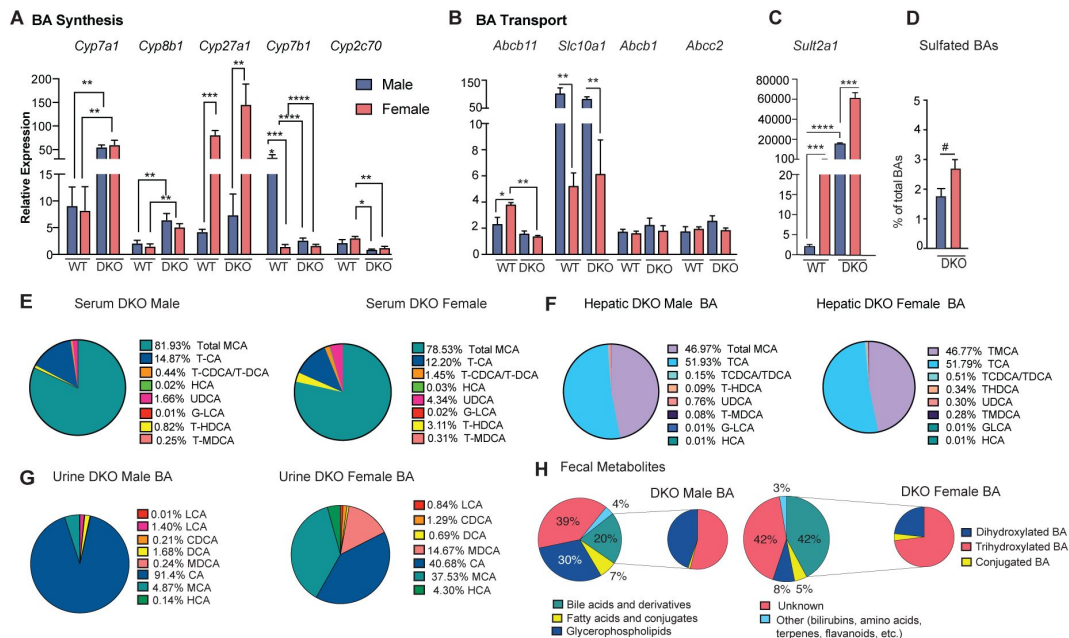


Figure 5

BA composition and metabolism are differentially regulated between the sexes of DKO mice.

(A) Hepatic mRNA expression of classical BA synthetic enzymes was elevated in DKO compared to WT mice. While the alternative BA synthesis encoding gene, *Cyp27a1*, was increased in females only. (B) Expression of hepatic BA transporters and (C) BA sulfotransferase in WT and DKO mice. (D) Percentages of sulfated BAs in DKO male and female serum (one-tailed t-test). (E-F) BA composition is slightly varied in serum, whereas it remains unchanged in the liver between DKO males and females. (G-H) BA composition in the urine was variable between the sexes, and BAs constitute a higher proportion of fecal metabolites in the DKO females compared to males ($n = 5-7/\text{group}$). Mean $\pm$ SEM; # $p < 0.05$, * $p < 0.01$, ** $p < 0.001$ compared to genotype or gender controls. One-way ANOVA with Bonferroni post hoc analysis was performed.

the males, whereas in the females, it was double the amount indicative of twice the amount being excreted in DKO females. These results indicate that female DKO mice may be protected against detrimental tumor-promoting BA signaling due to their higher BA excretion.

Increasing fecal BA excretion is sufficient to reduce liver cancer risk

Finally, to test this, we promoted fecal BA excretion in DKO males by using cholestyramine (CHR), a resin that binds BAs. We fed nine-month-old DKO male mice with a 2% CHR-containing diet since, by this age, tumor nodules have already developed. The CHR diet was continued until one year of age, mimicking a therapeutic intervention strategy (**Fig. 6A**). As expected, the CHR diet in DKO males led to a 50% reduction in circulating BA levels and altered BA composition (**Fig. 6B–C**). DKO males fed chow exhibited severe hepatic tumorigenesis, whereas CHR-fed DKO males had a drastically lower tumor burden with only small liver nodules and were protected from developing aggressive carcinomas. Histological analysis revealed that CHR treatment lowered the number of nodules and dysplastic changes but increased steatosis in DKO males (**Fig. 6D**). Conversely, increasing circulating BAs by causing biliary injury with 3,5-diethoxycarbonyl -1,4-dihydrocollidine (DDC) in DKO females resulted in the development of large liver tumors in DKO females (**Supplementary Fig. 7**).

Unbiased correlation analysis of hepatic and serum BA composition between the two cohorts of DKO mice revealed that the BA profiles of CHR-fed DKO male mice clustered with DKO females, whereas DDC-fed DKO female mice clustered with DKO males (**Supplementary Fig. 8**).

Overall, these findings demonstrate that modulating circulating BAs is sufficient to change the liver cancer outcome, wherein lowering their levels leads to subsequent protection and vice versa.

Discussion

Here, we demonstrate that the sex differences in BA homeostasis can contribute to the sexual disparity noted in HCC risk. Importantly, elevated BA concentrations are reported in patients with HCC [13, 26, 33–35, 59]. Using a genetic mouse model of excess BAs that develop spontaneous HCC, we uncovered distinct transcriptional control of metabolism between the two sexes. Both *Fxr* and *Shp* transcript levels were downregulated in HCC patients. Moreover, differential gene expression, specifically of the DKO female, correlated well with better survival, highlighting the translational relevance of our model. Thus, these gene signatures could be utilized as a potential prognostic marker for HCC progression and survival.

Both BA homeostasis and amino acid metabolism were altered between the two sexes. Of note, genes controlling ureagenesis were higher in the DKO females, and consistent with previous findings, we were able to recapitulate estrogen-mediated regulation of some of these genes signaling [48]. In line with these findings, patients with urea cycle enzyme deficiencies have a 200x higher incidence of HCC, highlighting the importance of amino acid metabolism in hepatic tumorigenesis [42–45]. Also, BAs have been shown to promote amino acid catabolic machinery [60], which indicates that BAs may be a central node in liver cancer. Intriguingly, hepatic urea analysis did not reveal any difference between the DKO male and female mice. A caveat being we measured a steady-state urea levels rather than the flux of this pathway.

We examined and found estrogen signaling can regulate the expression of BA synthesis and sulphation genes. DKO female mice challenged with the CA diet showed a robust increase in hepatic *Era* transcript, which coincided with the suppression in BA synthesis in the absence of *Fxr* and *Shp*. Consistently higher recruitment of ER to the classical BA synthetic genes was noted in

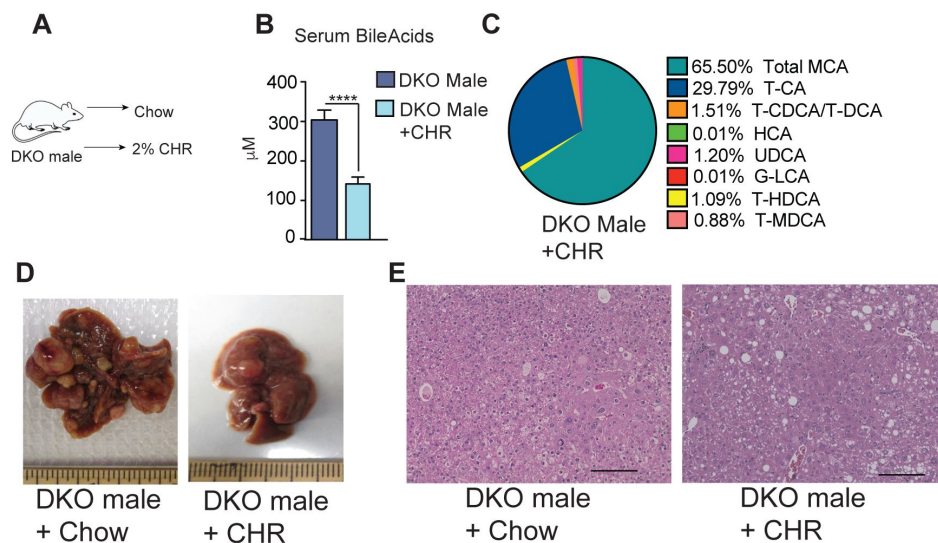


Figure 6

Treating with BA-binding resin reduces the tumor burden in DKO male mice.

(A) DKO male mice were fed a 2% cholestyramine (CHR)-enriched diet for 3 months until one year of age. (B-C) Serum BA levels and composition upon feeding DKO male mice CHR-enriched diet. (D) CHR dramatically reduced the HCC burden in DKO males. (E) Histological analysis shows HCC, bland tumor cells, and enlarged nuclei with irregular membranes in DKO male mice. CHR treatment results in smaller and fewer nodules but increases steatosis. ($n = 6-7/\text{group}$). Mean $\pm$ SEM; * $p < 0.01$; ** $p < 0.001$ compared to DKO controls.

female livers. DKO OVX mice with blunted *Era* gene expression, also showed lower transcript levels of *Cyp7a1* and *Cyp8b1* that were unaltered post-CA challenge. Currently, we do not know how to reconcile this data other than indicating a potential ER-independent mechanism.

Nonetheless, we confirmed the known sex differences in BA synthesis, such as a female-dominant *Cyp27a1* expression and male-dominant *Cyp7b1* pattern in WT mice. Loss of *Fxr* and *Shp* altered the expression of many genes irrespective of sex. For instance, *Cyp2c70* expression was reduced in both sexes, and the male dominance of *Cyp7b1* was lost in the DKO mice.

Of note, OVX of DKO females increased the serum BA levels and lost their protection against the development of liver tumorigenesis. This finding fully recapitulates the clinical data, wherein post-menopausal women are equally prone to HCC incidence as males.

BA analysis shows that DKO female mice have a hydrophilic composition and excrete BA proportions. So, we tested and demonstrated the potent therapeutic utility of reducing BA levels in serum using a generic FDA-approved BA binding resin, Cholestyramine (CHR), in dramatically reducing the tumor burden. This study highlights that lowering enterohepatic recirculation is a beneficial strategy in modulating liver cancer. Though *Cyp7a1* expression is reported to be induced in CHR-fed mice [61], long-term CHR feeding in DKO mice lowered *Cyp7a1* expression but induced *Cyp8b1* transcripts (**Supplementary Figure 8B**). Conversely, DDC-fed DKO females that develop hepatic tumors show a corresponding decrease in *Cyp8b1* transcript (**Supplementary Figure 8C**). Also, patients with HCC exhibit a reduction in *Cyp8b1* expression [62–64], which promotes a more hydrophilic ratio of BA composition.

Although species differences in BAs between mice and humans are a limitation, several fundamental understandings have been gained from mouse experiments. While this study demonstrates bile acids promote HCC progression, it does not investigate or provide evidence if excess bile acids are sufficient for HCC initiation. Another caveat is that the DKO mouse model mimics the progression of cholestasis to HCC and not all the etiologies, so the observed sex differences in circulating BAs may be limited to these subsets of HCC. Nevertheless, elevated BA concentrations are seen in various liver disease conditions and inherited disorders of cholestasis predispose to HCC onset. More recently, clinical studies support the utilization of BAs as prognostic markers [65–67].

Our findings demonstrate that female cholestatic mice exhibit increased excretion and lower serum BAs than males. However, hepatic BAs were not different between the sexes. These data highlight that circulating BAs contribute towards sex differences seen in HCC incidence. Accordingly, we show that lowering enterohepatic BA recirculation is beneficial in our model. Our results align with previous findings that had implicated intestinal FXR signaling as being crucial rather than hepatic FXR to prevent liver tumorigenesis [68]. Taken together, these results reveal that drugs inhibiting intestinal reabsorption of BAs (Asbt inhibitor, IBAT inhibitor) that are on clinical trials for NASH and cholestasis can be evaluated as potential therapeutics to combat HCC.

Methods

Experimental Design

This study was designed to elucidate the role of bile acids (BAs) in the sexually dimorphic incidence of HCC and assess the therapeutic benefits of reducing circulating BA levels on HCC development. *Fxr*^{-/-}, *Shp*^{-/-} (DKO) mice were maintained on a C57BL/6 background at the University of Illinois, Urbana-Champaign. Male and female mice were sacrificed at 8–12 weeks or 6 and 12 months after birth. Mice were housed on a standard 12-hour light/dark cycle and fed normal chow and water ad libitum. In order to study estrogen signaling, bilateral ovariectomies were

performed on WT and DKO females at 8-10 weeks old, followed by one week of recovery. 2% Cholestyramine-supplemented chow was fed to 9-month-old DKO male mice for a period of 3 months while 0.1% DDC (3,5-diethoxycarbonyl-1, 4-dihydrocollidine) was fed to 10-month-old DKO female mice for 3 months. Urea cycle studies were performed on mice after overnight fasting. All studies were carried out as outlined in the Guide for the Care and Use of Laboratory Animals prepared by the National Academy of Sciences and published by the National Institutes of Health (National Institutes of Health publication 86-23, revised 1985). For biological harvesting, mice were anaesthetized and euthanized as described by IUCAC. Tissue was flash-frozen in liquid nitrogen and blood was collected and spun down for serum.

Serum Chemistry

Blood was collected by retro-orbital bleeding and centrifuged at 8000g x 10 minutes to separate serum. Serum ALT and AST were measured using Infinity ALT and Infinity AST kits (Thermo Scientific). Calorimetric measurement of serum and hepatic bile acids was performed with the Total Bile Acid (NBT method) kit (Genway Biotech).

Bile Acid analysis

Serum and urine from DKO male and female mice fed chow were analyzed for the composition of bile acids and their sulfated metabolites at the University of Nebraska Medical Center. Liquid chromatographic-mass spectrometry analysis was performed with a Waters ACQUITY column (Milford, MA). Bile acids and internal standards were prepared in methanol and analyzed. Serum from DKO male and female mice fed chow, DKO males fed CHR, and DKO females fed DDC was analyzed for bile acid composition at Baylor College of Medicine Metabolomics Core, Houston, Texas. Liquid chromatographic-mass spectrometry analysis was performed with a Waters ACQUITY UPLC BEH C18 column (Milford, MA). Bile acids were detected in negative mode, with L-Zeatin added to each sample as an internal standard.

Metabolite Profiling

Liver tissue was weighed and sonicated in 70% methanol, followed by centrifugation. The supernatant was flash-frozen and used for subsequent LC-MS analysis for urea cycle metabolites. Tissue lysate was used for the BCA assay to determine the protein concentration of each sample. All metabolite concentrations were normalized to a protein concentration of the lysate.

Untargeted Metabolomics

Fecal samples were weighed into microcentrifuge tubes and homogenized in 50% MeOH/H₂O solution with a 1:10 w/v ratio, for 5 minutes at 5 Hz. The samples were centrifuged at 14000 rpm for 15 minutes, then a 200uL aliquot of each supernatant was transferred to a 96-well plate and dried under centrifugal vacuum. The dried extracts were covered and stored at -80 °C until analysis, at which time the samples were resuspended in 200uL of 50% MeOH/H₂O solution with 1uM sulfadimethoxine as internal standard and diluted three-fold for analysis. Untargeted LC-MS/MS was performed on a Thermo Vanquish UPLC system coupled to a Q-Exactive Orbitrap mass spectrometer (ThermoFisher Scientific, Bremen, Germany). A polar C18 column (Kinetex polar C18, 100 x 2.1 mm, 2.6 µm particle size, 100 Å pore size; Phenomenex, Torrance, CA USA) was used as the stationary phase, and a high-pressure binary gradient pump was used to deliver the mobile phase, which consisted of solvent A (100% H₂O + 0.1 % formic acid (FA)) and solvent B (100% acetonitrile (ACN) + 0.1 % FA). The flow rate was set to 0.5mL/min and the injection volume for each sample was 5uL. Following injection, samples were eluted with the following gradient: 0-1.0 min, 5% B; 1.0-1.1 min, 25%; 6.0 min, 70%; 7.0 min, 100%; 7.5-8.0 min, 5%. MS data was collected in positive mode and electrospray ionization (ESI) parameters were set to 53 L/min for sheath gas, 14 L/min for auxiliary gas, 0 L/min for spare gas, and 400°C for auxiliary gas temperature. The spray voltage was set to 3500 V, the capillary temperature to 320 °C, and the S-Lens radio frequency level to 50 V. MS1 data were collected from 150-1500 m/z with a resolution of 35,000 at m/z 200 with one

micro scan. The maximum ion injection time was set to 100 ms with an automatic gain control (AGC) target of 1.0E6. MS/MS spectra were collected using data dependent acquisition (DDA), where the top 5 most abundant ions in the MS1 scan were selected for fragmentation. Normalized collision energies were increased stepwise from 20, 30, to 40. MS2 data were collected with a resolution of 17,500 at m/z 200 with one micro scan and an AGC of 5.0E5. All untargeted LC-MS/MS data are publicly available from the MassIVE data repository under accession number MSV000089715. MS1 feature detection and MS/MS pairing was performed using MZmine 2.37corr17.7_kai_merge (55, 56). An intensity threshold of 5E4 and 1E3 were set for MS1 and MS2 detection, respectively, with centroid data. MS1 chromatogram construction was performed using the ADAP chromatogram builder, where the minimum group size was set to 5, group intensity threshold was 5E4, minimum highest intensity was 1.5E5, and mass tolerance was 0.005 m/z or 10 ppm. Chromatogram deconvolution was then performed using a local minimum search algorithm with a chromatographic threshold of 80%, a search minimum in retention time (RT) range of 0.2 min, minimum relative height of 1%, minimum absolute threshold height of 1.5E5, minimum ratio for top/edge of 1, and a peak duration of 0.05-2.0 min. Pairing between MS1 and MS2 was performed with a mass tolerance of 0.005 m/z or 10 ppm and RT range of 0.2 min. Isotope peaks were grouped, then features from different samples were aligned using the same mass and RT tolerances; alignment was performed by placing a weight of 75 on m/z and 25 on RT. A peak area feature table was exported as a .csv file and consensus MS/MS spectral data were exported in mgf format. Feature-based molecular networking and MolNetEnhancer workflows were then performed with this data using GNPS (gnps.ucsd.edu). The corresponding jobs can be found at: <https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=d697d44ec18440d29d0771f84ba7cccdand> and <https://gnps.ucsd.edu/ProteoSAFe/status.jsp?task=e3edba56efba4a27b073a9031c60b5e5>, respectively.

Histology

Liver samples were fixed in 10% neutral-buffered formalin, sections were cut at a thickness of ~5 microns and used for hematoxylin and eosin, and sirius red staining.

RNA extraction and quantitative PCR analysis

Total RNA from the liver was prepared according to the TRIzol (Invitrogen) protocol. cDNA was synthesized using Maxima Reverse Transcriptase (Thermo Scientific) according to the manufacturer's protocol. q-RT-PCR was performed on an Illumina Eco Platform. For qRT-PCR analysis, 50 ng of cDNA was added to each SYBR green-based reaction. qRT-PCR primers are provided in **Supplementary Table 7**.

Microarray

Microarray was performed by Dr. Ju-Seog Lee's laboratory at the MD Anderson Cancer Center. Liver samples from 12-month-old male and female WT and DKO mice were collected and snap-frozen. Total RNA was isolated, labeled, and hybridized to BeadChip Array MouseWG-6 (Illumina). Bead chips were scanned with an Illumina BeadArray Reader. Microarray analysis was performed on the Illumina mouseRefseq-8 Expression platform. Upregulated gene sets were generated from genes with fold change > 1.3 ($p < 0.0001$) compared to the control group (i.e. DKO males vs. DKO females). These gene sets were then used for downstream analyses with DAVID Bioinformatics Resources Analysis Software and ClueGO [36].

Transcription Factor Motif Analysis

GeneXplain software was used to identify enriched transcription factor binding sites (TFBS) using the upregulated gene sets generated from the microarray. The analysis included regions from -1000 to 100 bp relative to the transcription start site. Only those TFBS enriched with $p \leq 0.01$ were included in the tables.

Extraction of transcriptomic signature

Multiple transcriptomic signatures were extracted from the microarray data of the DKO mouse model (Supplementary Table 1). DKO_All signature was generated from the comparison between wild type (WT) male and female mice, and DKO_Male and DKO_Female signatures from WT male and female mice, respectively. DKO_FvsM, DKO_Estrogen, DKO_BA, and DKO_Urea signatures were made from the comparison between DKO male and female mice. Signature genes were selected by T-test and logFC ($p < 0.001$ and $\log_2FC > 1$ or < -1) using the gene expression dataset after normalization.

Gene expression data from HCC tumors

Gene expression data from the National Cancer Institute (NCI) cohort were generated in earlier studies [69–71], and the data are publicly available from the NCBI's GEO database (GSE1898 and GSE4024). Gene expression data from Korea, Samsung, Modena, and Fudan cohorts have been described previously and are available from the NCBI's GEO database (accession numbers, GSE14520, GSE16757, GSE43619, GSE36376, and GSE54236) [72–76]. TCGA RNA sequencing data for HCC was downloaded from the University of California, Santa Cruz, Genomics Institute (<https://xenabrowser.net/>) [12]. FPKM-normalized data were log-transformed.

Tumor specimens and clinical data were obtained from HCC patients who had undergone hepatectomy as a primary treatment for HCC at multiple institutes, as described in their original study. Except for the TCGA cohort, patients and tissues were collected based on the availability of high quality of frozen tissues for genomic studies. For TCGA cohort [77], surgical resection of biopsy biospecimens were collected from patients diagnosed with HCC and had not received prior treatment for their disease (ablation, chemotherapy, or radiotherapy). Institutional review boards at each tissue source site reviewed protocols and consent documentation and approved the submission of cases to TCGA. Hematoxylin and eosin (H&E) stained sample were subjected to independent pathology review to confirm that the tumor specimen was histologically consistent with the allowable HCC. Each case was reviewed independently by at least 3 liver pathologists, with no clinical or molecular information.

Prediction model

To predict the class similar to the DKO signature in the human HCC cohort, we used a classification algorithm based on Bayesian compound covariate predictor (BCCP). After the integration of the signature matrix and the human HCC dataset, the Bayesian probability for each human HCC sample were calculated by using the class prediction procedure in BRB Arraytools. We set 0.5 as the cut-off of Bayesian probability for each signature.

Validation of clinical relevance

Prognostic significance was evaluated rigorously for overall and recurrence-free survival in the human HCC cohort based on the predicted class calculated by the BCCP algorithm using multiple DKOsignatures. A total of 5 human HCC transcriptomic cohorts were used in this study (Fudan, Korea, Samsung, TCGA, Modena). All DKOsignatures were evaluated in each human HCC cohort and meta-cohort. To identify the gender difference in the human HCC cohort, we did subgroup analysis for gender and age in the meta-cohort. BCCP scores (BCCP probability) were compared in all populations and gender subgroups. The analysis for potential correlation between the class predicted by DKO-signature and staging HCC in terms of TNM, BCLC, and CLIP classification was performed in the meta-cohort.

ERα ChIP-Seq Analysis

ERα-ChIP assay was performed in both sexes of WT and DKO mice. ERα-F10 antibody (sc-8002, Santa Cruz) was used to perform the pulldown followed by qPCR. We also analyzed BED files for ERα ChIP-Seq from three independent studies [78–80], which were obtained from Cistrome DB and visualized using the UCSC genome browser on mouse GRCm38/mm10 assembly. For tracks 3 and 4, the track size was set to auto-size.

Statistical Analysis

All statistical tests were performed using GraphPad Prism software. Data are presented as means ± SEM. Multiple group comparisons were analyzed using one-way and two-way ANOVA with the post hoc Bonferroni test. Unpaired t-test was used for comparison between two groups. P values ≤0.05 were determined to be significant unless otherwise noted in legends.

Supplementary Table 1: List of genes in different DKO gene signature category used for analysis. See Excel sheet under Additional files.

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Additional information

Data Availability

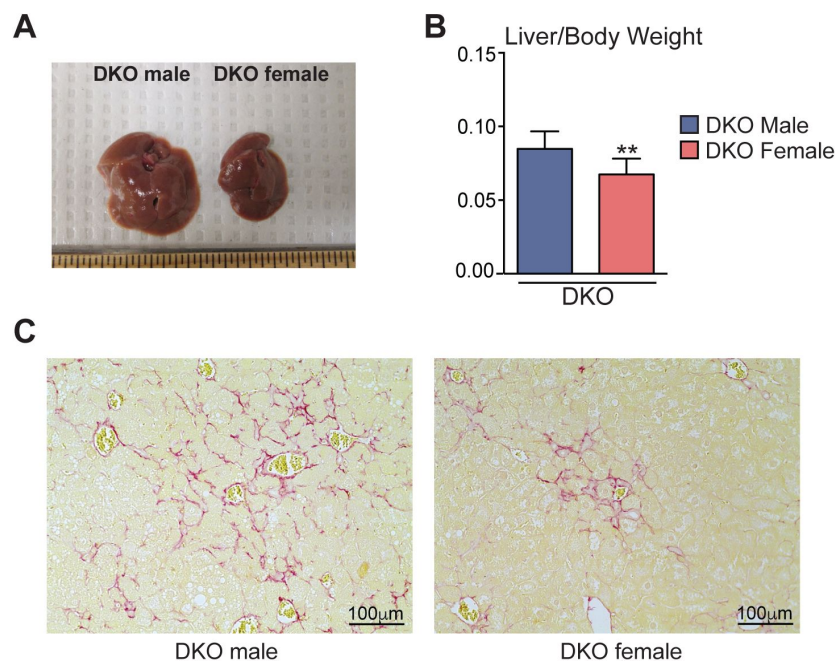
The gene expression data generated and used in this publication have been deposited in NCBI's Gene Expression Omnibus and are accessible through GEO Series accession number GSE151524. (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE151524>).

Author contributions

M.E.P., S.K., A.E.D and S.A. contributed to conception, experimental design, data acquisition, analysis and interpretation, and drafting the article. L.J.T. helped with urea cycle expression analysis, R.N.T. and Y.A. analyzed bile acid composition and interpretation from serum and urine, E.G., M.P, P.D performed untargeted metabolomics, A.E.D, Q.Z, Z.M.E analyzed ERα ChIP-seq data, H.S.L and J.S.L performed all the HCC patient cohort analysis, M.J.F. evaluated and interpreted the liver histology.

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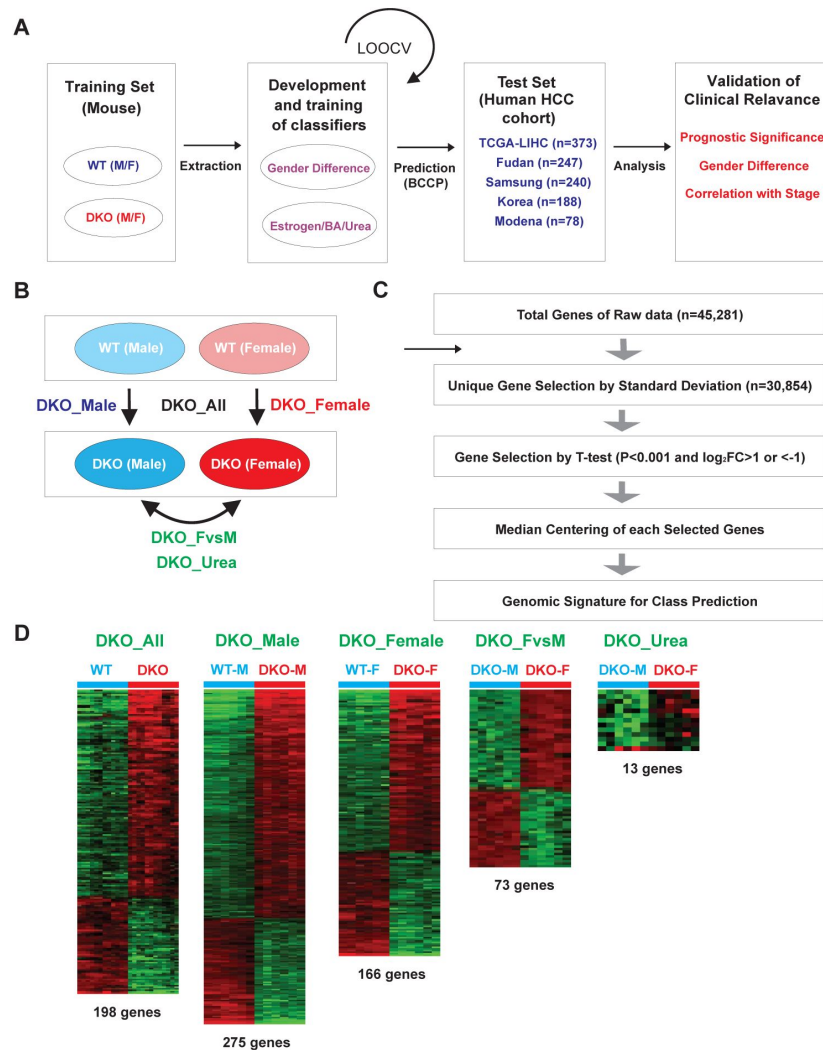
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Supplementary Fig. 1

DKO female mice exhibit reduced liver injury.

(A-B) Compared to DKO males, the female mice showed smaller liver size and liver-to-body weight ratio. (C) Sirius red staining revealed less collagen staining (red) in DKO females. (Six month old mice, $n = 6/\text{group}$). Mean $\pm$ SEM. * $p < 0.01$, ** $p < 0.001$ compared to genotype or gender controls. Unpaired t-test was used to analyze the data.



Supplementary Fig. 2.

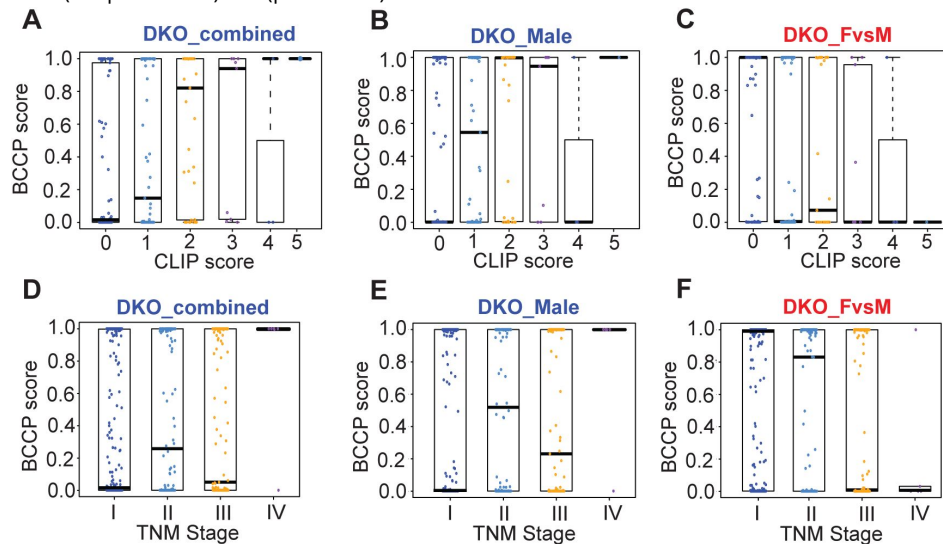
Schematic of the pipeline used to obtain gene signatures to predict outcomes in HCC patient cohorts.

(A) Workflow schematic of the gene expression analysis. (B-C) Different gene signatures were extracted from the microarray data by comparing wild-type (WT) male and female gene expression patterns with that of DKO male and DKO female, respectively. These gene lists were used to predict overall survival and recurrence-free survival using five human clinical cohorts. (D) Gene changes in these defined sets DKO males (DKO_M), DKO females (DKO_F), or combined (DKO_ALL), or DKO_F vs M (DKO female gene signature that does not overlap with DKO males), DKO_Estrogen, DKO_BA, and DKO_Urea are shown. These genes were selected by t-test and $\log_2$ Fold Change ($p < 0.001$ and $\log_2 FC > 1$ or < -1).

Supplementary Fig. 3

Correlation of DKO gene signatures with various clinical stages of HCC.

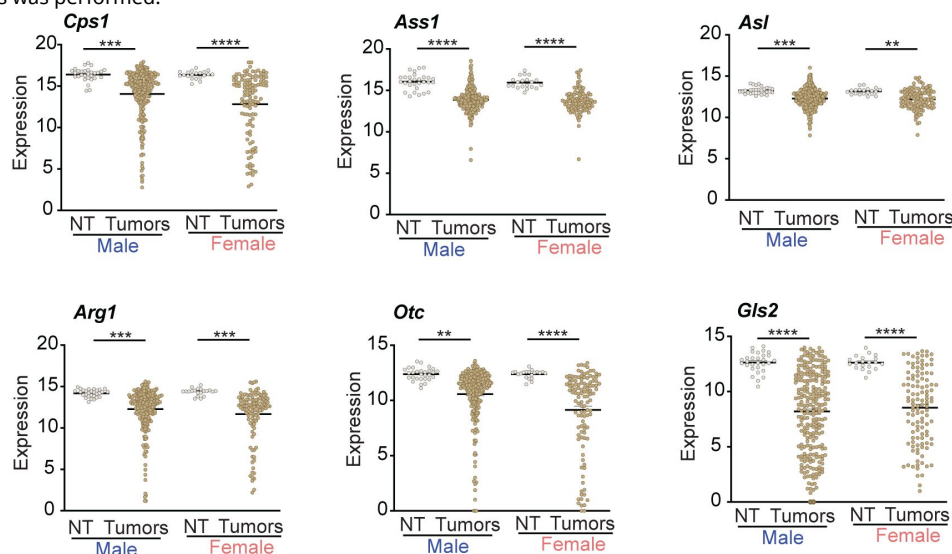
Clinical relevance of combined gene changes in DKO male and female (DKO_Combined), or DKO male-specific, and the distinct gene set of DKO female vs. male (DKO_FvsM) were tested in five HCC cohorts. The trained BCCP (Bayesian compound covariate predictor) algorithm using the three DKO gene sets was tested across the HCC stages as determined by (A-C) CLIP (Cancer of the Liver Italian Program) or (D-F) TNM (Tumor, Node, and Metastasis). The probability of gene signatures was generated from 0 (not predictable) to 1 (predictable).



Supplementary Fig. 4

Expression of the urea cycle genes are reduced in human liver cancer.

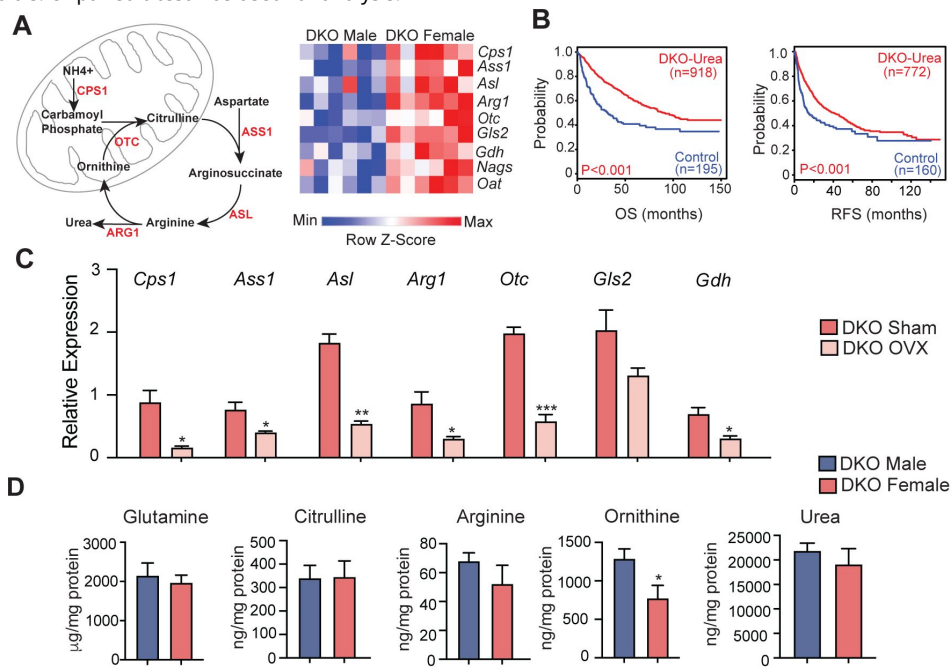
RNA-seq from TCGA-LIHC (The Cancer Genome Atlas-Liver Hepatocellular Carcinoma Collection) database was analyzed. Several genes that encode enzymes involved in ureagenesis exhibited a reduction in their transcript levels in tumors compared to non-tumor (NT) tissue. (Males: n=29 NT, n=245 tumors; Females: n=20 NT and n=114 tumors). Mean $\pm$ SEM; * p <0.01, ** p <0.001, *** p <0.0001 compared to their respective sex-specific controls. One-way ANOVA with Bonferroni post hoc analysis was performed.



Supplementary Fig. 5

Urea cycle genes elevated in DKO females correlate with better patient survival and reduced after ovariectomy.

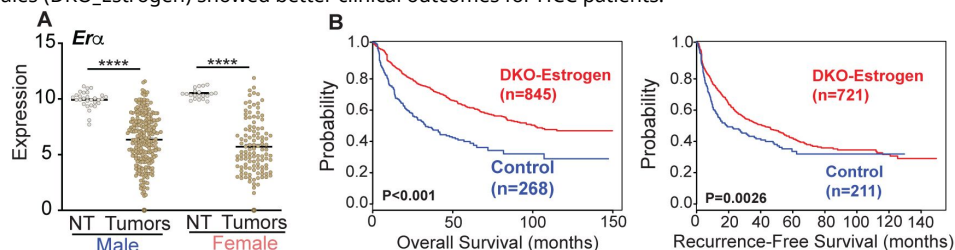
(A) Schematic of the urea cycle, with a representative heat map of urea cycle genes from the microarray showing higher expression in DKO female livers ($n = 6/\text{group}$). (B) Survival curves from HCC patients who exhibit increased expression of urea cycle genes. (C) Expression of urea cycle genes was reduced in DKO female mice following OVX ($n = 4\text{-}5/\text{group}$). and (D) LC-MS quantification of urea cycle intermediates ($n = 4\text{-}5/\text{group}$). Mean $\pm$ SEM; * $p < 0.01$, ** $p < 0.001$ compared to genotype or gender controls. Unpaired t-test was used for analysis.



Supplementary Fig. 6

Estrogen receptor signaling positively correlates with better survival in HCC clinical samples.

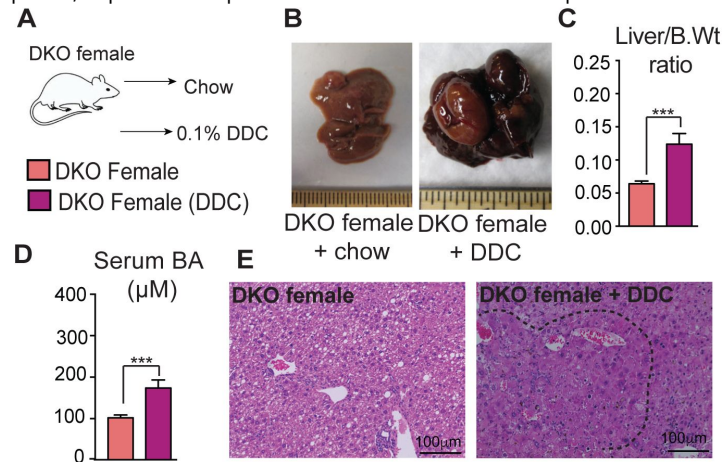
(A) RNA-seq from TCGA-LIHC (The Cancer Genome Atlas-Liver Hepatocellular Carcinoma Collection) data was analyzed. *Era* transcript levels were reduced in tumors compared to non-tumor (NT) tissue. (Males: $n = 29$ NT, $n = 245$ tumors; Females: $n = 20$ NT and $n = 114$ tumors). Mean $\pm$ SEM; * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$ compared to their respective sex-specific controls. One-way ANOVA with Bonferroni post hoc analysis was performed. (B) Kaplan Meier Survival graphs using *Era* targets altered in DKO females (DKO_Estrogen) showed better clinical outcomes for HCC patients.



Supplementary Fig. 7

DKO females challenged with the DDC diet developed liver tumors.

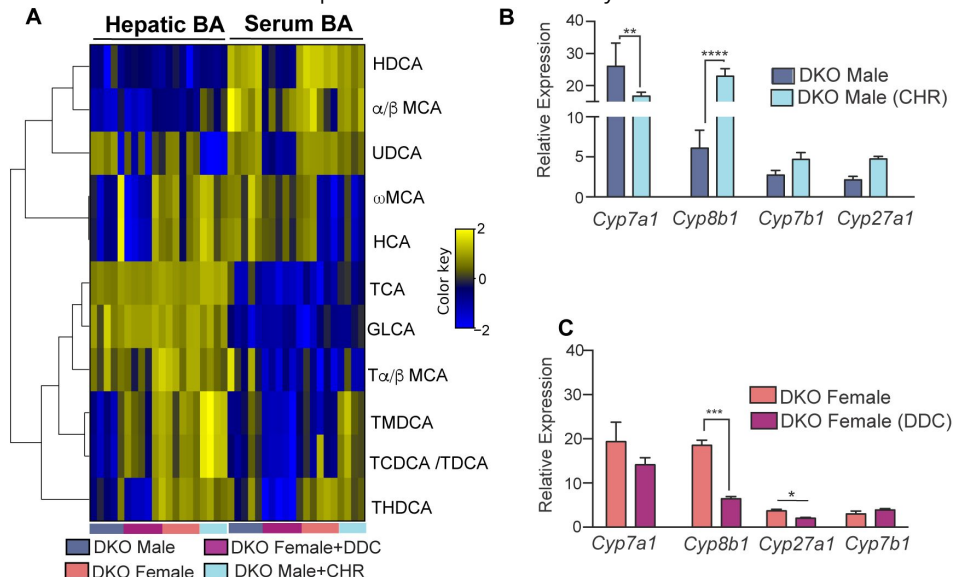
(A-B) DKO females were fed a 0.1% DDC diet, as shown in the schematic, for 5 months, and this treatment led to visible tumors at one year. (C-D) The diet elevated liver-to-body weight ratio and serum BAs in DKO females. (E) Histology revealed cholangitis, ductular reaction, and tumor nodules (marked by a dotted line) in DKO female livers+ DDC treatment. ($n = 4-7/\text{group}$). Mean $\pm$ SEM; * $p < 0.01$; ** $p < 0.001$ compared to chow-fed DKO controls. Unpaired t-test was used for analysis.



Supplementary Fig. 8

BA composition and expression of BA synthesis genes in DKO mice after different treatments.

(A) Correlation analysis of serum and hepatic composition of BAs post-DDC diet in DKO female mice ($n = 5-8$ mice/group) and post-CHR diet in DKO male mice ($n = 4-5$ mice/group) along with their sex-specific chow controls. (B-C) Relative expression of classical and alternative BA synthesis genes after CHR and DDC treatments. Mean $\pm$ SEM; * $p < 0.01$; ** $p < 0.001$, *** $p < 0.0001$ compared to their chow-fed DKO controls. Unpaired t-test was used for analysis.



Supplementary Table 2.

Transcription Motifs Enriched in DKO male livers compared to DKO females.

TF Name	Density ratio	P-Value
PPAR γ	1.30126	6.3527E-11
TCF3	1.33458	7.2023E-11
TCF4	1.27146	5.5608E-9
p73	1.30219	1.1841E-5
LRH1	1.1917	2.3061E-5
AR	1.3065	2.7733E-5
FOXA2	1.21861	6.526E-4
MEF2A	1.35309	0.00153
MEF2C	1.34771	0.0016
TBR2 (EOMES)	1.15298	0.00168
TR4	1.45287	0.00195
GATA3	1.95407	0.00351
c-JUN	1.20405	0.00491
STAT1	1.17336	0.00505
RXR α	1.42981	0.00512
STAT5a	1.19101	0.00557
FOXA1	1.53342	0.00674
STAT4	1.42981	0.00707
NRF2	3.06387	0.00959
KLF4	1.08893	0.01000

Supplementary Table 3.

Transcription Motifs Enriched in DKO female livers compared to DKO males.

TF Name	Density ratio	P-Value
USF1	3.10842	6.0846E-4
GATA1	1.26352	0.00179
THAP1	3.38041	0.00226
FOXO1	2.6577	0.00247
E2F4	1.53201	0.00262
SP1	6.29455	0.00267
E2F6	1.79844	0.00439
GR	1.31386	0.00484
ZFX	1.2357	0.00614
NKX2-5	1.72517	0.00633
USF2	2.09818	0.00675
ZBTB33	1.76247	0.0091
YY1	1.29119	0.01000
ESR1	1.72159	0.01000

Supplementary Table 4.

Summary of HCC gene expression data sets

Cohorts	Platform	Number of patients	Source
Fudan	Affymetrix U133A2	247	GEO, GSE14520
Samsung	RNA-seq Illumina HumanHT-12 V4.0	240	GEO, GSE36376
TCGA -LIHC	RNA seq	373	https://xenabrowser.net/
Korea	Illumina V2, V4	188	GEO, GSE16757, GSE43619
Modena	Agilent-014850	78	GEO, GSE54236

Supplementary Table 5

Serum BA composition in DKO mice.

	DKO Male n=4	DKO Female n=3
Serum		
Amidation		
% Amidation	73.67%	77.66%
% G-Amidation	0.14%	0.11%
% T-Amidation	73.53%	77.55%
BA Hydroxylation		
% Mono-OH	0.15%	0.27%
% Di-OH	13.53%	12.19%
% Tri-OH	81.78%	82.51%
BA Composition		
% LCA	0.15%	0.27%
% UDCA	0.39%	0.47%
% CDCA	0.79%	1.01%
% DCA	12.09%	10.41%
% HDCA	0.19%	0.23%
% MDCA	0.07%	0.06%
% CA	69.56%	65.50%
% MCA	12.21%	17.00%
% HCA	0.01%	0.00%
Primary BA (μM)	60.10	36.27
Secondary BA (μM)	12.68	7.39
Primary/Secondary	5.62	5.36
12α/non12α	5.22	4.03
% Sulfation	1.72%	2.66%
Hydrophobicity Index	0.029	-0.029

	DKO Male	DKO Female
Urine	n=4	n=3
Amidation		
% Amidation	79.62%	20.52%
% G-amidation	0.20%	0.52%
% T-amidation	93.32%	37.81%
BA Hydroxylation		
% Mono-OH	0.01%	0.78%
% Di-OH	3.51%	15.48%
% Tri-OH	96.17%	76.71%
BA Composition		
% LCA	0.01%	0.78%
% UDCA	1.40%	0.00%
% CDCA	0.21%	1.20%
% DCA	1.67%	0.64%
% HDCA	0.00%	0.00%
% MDCA	0.24%	13.64%
% CA	91.17%	37.82%
% MCA	4.86%	34.90%
% HCA	0.14%	4.00%
% Primary BA	96.36%	80.00%
% Secondary BA	3.64%	20.00%
Primary/Secondary	33.52	4.13
12a/non12a	2.21	0.63
% Sulfation	79.62%	20.52%
Hydrophobicity Index	-0.09	-0.13

Supplementary Table 6.

Urine BA composition in DKO mice.

Gene Name	Forward Primer 5' to 3'	Reverse Primer 5' to 3'
<i>36B4</i>	AGATGCAGCAGATCCGCAT	GTTCTTGCCCATCAGCACC
<i>Ccnd1</i>	CCTCCTTCTGCACACATTTGA	GCTGCGAAGTGGAACCATC
<i>Cyp7a1</i>	CAGGGAGATGCTCTGTGTTCA	AGGCATACATCCCTTCCGTGA
<i>Cyp8b1</i>	AAGGCTGGCTTCCTGAGCTT	AACAGCTCATCGGCCTCATC
<i>Cyp27a1</i>	CCTACATCCATTCCGGCTCT	CCAGGGCAATCTCATACTTC
<i>Cyp7b1</i>	GACGATCCTGAAATAGGAGCACA	AATGGTGTGTTGCTAGAGAGGCC
<i>Cyp2c70</i>	TGGCTTTCTCAGCAGGAAGAA	AACTGGCTTGGTGTGATGT
<i>Abcb11</i>	CACACAAAGCCCCTACCAGT	CCAAGAGGCAGCTATCAGGC
<i>Slc10a1</i>	GGTGCCCTACAAAGGCATTA	GTTGCCACATTGATGACAG
<i>Abcb1</i>	ACTCGGGCGCAGAACTTTGA	GCACCAAAGACAACAGCAGA
<i>Abcc2</i>	GCACTGTAGGCTCTGGGAAG	CATTTCCAAGTCTGGGAGGA
<i>Abcc3</i>	GCAGAGACAGGCAATGTGAA	GAAAGCTGACAGCATGACCA
<i>Sult2a1</i>	CCTCAAAAGAAATGTTCTAATCGGA	CTTTCATGGCTTGGAAAGAGCTGTA
<i>Esr1</i>	GACAAGCGGCGTAAAAG	CATTATGGGGTCTGGTCCTG
<i>Cps1</i>	AGGCACCACCATTACCTCAG	CGGGTTCATCAGGACTGTTT
<i>Ass1</i>	ACACCTCCTGCATCCTCGT	GCTCACATCCTCAATGAACACCT
<i>Asl</i>	CTAGAGGTACAGAAGCGG	TGCTGTTGAGAGTGATGG
<i>Arg1</i>	CTCCAAGCCAAAGTCCTTAGAG	AGGAGCTGTCATTAGGGACATC
<i>Otc</i>	TTTGGGTGTGAATGAAAGTCTC	TGATGATTGGGATGGATGCT
<i>Gls2</i>	CCGTGGTGAACCTGCTATTT	TGCGGGAATCATAGTCCTTC
<i>Gdh</i>	GGCCGATTGACCTTCAAATA	TCCTGTCCTGGAACCTCTGCT
<i>Glul</i>	CAGGCTGCCATACCAACTTCA	TCCTCAATGCACTTCAGACCAT
ChIP-PCR		
<i>Cyp7a1</i>	TTCCCCATCAGTTTGCAGGT	ACCTCCCAAATCTGCGTCTT
<i>Cyp8b1</i>	GAGGACCCAGAAACAGGTGG	TGTCTACTCCCTACTGGGGC
<i>Ldlr</i>	CAGAGCCATCGTAGTGGACC	TTGGTCTATCACCGACACCC
<i>Pgr</i>	TGATGTGGTCTATGCAGGGC	TGGTGCTCTTTACATGGTAGTTG

Supplementary Table 7.

Primer sequences used

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American Cancer Society (ACS 132640-RSG)

Additional files

Supplemental Table 1. [🔗](#)

Supplementary Tables. [🔗](#)

References

1. El-Serag H.B (2012) **Epidemiology of viral hepatitis and hepatocellular carcinoma** *Gastroenterology* **142**:1264–1273 [Google Scholar](#)
2. White D.L., et al. (2017) **Incidence of Hepatocellular Carcinoma in All 50 United States, From 2000 Through 2012** *Gastroenterology* **152**:812–820 [Google Scholar](#)
3. Mortality G.B.D. (2015) **C. Causes of Death, Global, regional, and national age-sex specific all-cause and cause-specific mortality for 240 causes of death, 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013.** *Lancet* **385**:117–71 [Google Scholar](#)
4. Forner A., Llovet J.M., Bruix J. (2012) **Hepatocellular carcinoma** *Lancet* **379**:1245–55 [Google Scholar](#)
5. Llovet J.M., et al. (2021) **Hepatocellular carcinoma** *Nat Rev Dis Primers* **7**:6 [Google Scholar](#)
6. Liu P., et al. (2017) **Age-specific sex difference in the incidence of hepatocellular carcinoma in the United States** *Oncotarget* **8**:68131–68137 [Google Scholar](#)
7. Yu M.W., et al. (2003) **Role of reproductive factors in hepatocellular carcinoma: Impact on hepatitis B- and C-related risk** *Hepatology* **38**:1393–400 [Google Scholar](#)
8. Ma W.L., et al. (2008) **Androgen receptor is a new potential therapeutic target for the treatment of hepatocellular carcinoma** *Gastroenterology* **135**:947–55 [Google Scholar](#)
9. McGlynn K.A., et al. (2015) **Reproductive factors, exogenous hormone use and risk of hepatocellular carcinoma among US women: results from the Liver Cancer Pooling Project** *Br J Cancer* **112**:1266–72 [Google Scholar](#)
10. Li Z., et al. (2012) **Foxa1 and Foxa2 are essential for sexual dimorphism in liver cancer** *Cell* **148**:72–83 [Google Scholar](#)
11. Naugler W.E., et al. (2007) **Gender disparity in liver cancer due to sex differences in MyD88-dependent IL-6 production** *Science* **317**:121–4 [Google Scholar](#)
12. Pavlova N.N., Thompson C.B. (2016) **The Emerging Hallmarks of Cancer Metabolism** *Cell Metab* **23**:27–47 [Google Scholar](#)
13. Anakk S., et al. (2013) **Bile acids activate YAP to promote liver carcinogenesis** *Cell Rep* **5**:1060–9 [Google Scholar](#)
14. Kim I., et al. (2007) **Spontaneous hepatocarcinogenesis in farnesoid X receptor-null mice** *Carcinogenesis* **28**:940–6 [Google Scholar](#)
15. Zhang Y., et al. (2008) **Orphan receptor small heterodimer partner suppresses tumorigenesis by modulating cyclin D1 expression and cellular proliferation** *Hepatology* **48**:289–98 [Google Scholar](#)

16. Yang F., et al. (2007) **Spontaneous development of liver tumors in the absence of the bile acid receptor farnesoid X receptor** *Cancer Res* **67**:863–7 [Google Scholar](#)
17. Gomez-Ospina N., et al. (2016) **Mutations in the nuclear bile acid receptor FXR cause progressive familial intrahepatic cholestasis** *Nat Commun* **7**:10713 [Google Scholar](#)
18. Van Mil S.W., et al. (2007) **Functional variants of the central bile acid sensor FXR identified in intrahepatic cholestasis of pregnancy** *Gastroenterology* **133**:507–16 [Google Scholar](#)
19. Nishigori H., et al. (2001) **Mutations in the small heterodimer partner gene are associated with mild obesity in Japanese subjects** *Proc Natl Acad Sci U S A* **98**:575–80 [Google Scholar](#)
20. Kong B., et al. (2009) **Farnesoid X receptor deficiency induces nonalcoholic steatohepatitis in low-density lipoprotein receptor-knockout mice fed a high-fat diet** *J Pharmacol Exp Ther* **328**:116–22 [Google Scholar](#)
21. Wolfe A., et al. (2011) **Increased activation of the Wnt/beta-catenin pathway in spontaneous hepatocellular carcinoma observed in farnesoid X receptor knockout mice** *J Pharmacol Exp Ther* **338**:12–21 [Google Scholar](#)
22. Park Y.Y., Choi H.S., Lee J.S. (2010) **Systems-level analysis of gene expression data revealed NR0B2/SHP as potential tumor suppressor in human liver cancer** *Mol Cells* **30**:485–91 [Google Scholar](#)
23. He N., et al. (2008) **Epigenetic inhibition of nuclear receptor small heterodimer partner is associated with and regulates hepatocellular carcinoma growth** *Gastroenterology* **134**:793–802 [Google Scholar](#)
24. Eaton J.E., et al. (2013) **Pathogenesis of primary sclerosing cholangitis and advances in diagnosis and management** *Gastroenterology* **145**:521–36 [Google Scholar](#)
25. Strautnieks S.S., et al. (2008) **Severe bile salt export pump deficiency: 82 different ABCB11 mutations in 109 families** *Gastroenterology* **134**:1203–14 [Google Scholar](#)
26. van Wessel D.B.E., et al. (2020) **Genotype correlates with the natural history of severe bile salt export pump deficiency** *J Hepatol* **73**:84–93 [Google Scholar](#)
27. Parks D.J., et al. (1999) **Bile acids: natural ligands for an orphan nuclear receptor** *Science* **284**:1365–8 [Google Scholar](#)
28. Russell D.W (2003) **The enzymes, regulation, and genetics of bile acid synthesis** *Annu Rev Biochem* **72**:137–74 [Google Scholar](#)
29. Thomas C., et al. (2008) **Targeting bile-acid signalling for metabolic diseases** *Nat Rev Drug Discov* **7**:678–93 [Google Scholar](#)
30. Wang H., et al. (1999) **Endogenous bile acids are ligands for the nuclear receptor FXR/BAR** *Mol Cell* **3**:543–53 [Google Scholar](#)
31. Wang L., Soroka C.J., Boyer J.L. (2002) **The role of bile salt export pump mutations in progressive familial intrahepatic cholestasis type II** *J Clin Invest* **110**:965–72 [Google Scholar](#)
32. Anakk S., et al. (2011) **Combined deletion of Fxr and Shp in mice induces Cyp17a1 and results in juvenile onset cholestasis** *J Clin Invest* **121**:86–95 [Google Scholar](#)

33. Bernstein H., et al. (2009) **Bile acids as endogenous etiologic agents in gastrointestinal cancer** *World J Gastroenterol* **15**:3329–40 [Google Scholar](#)
34. Sun L., et al. (2016) **Bile acids promote diethylnitrosamine-induced hepatocellular carcinoma via increased inflammatory signaling** *Am J Physiol Gastrointest Liver Physiol* **311**:G91–G104 [Google Scholar](#)
35. Xie G., et al. (2016) **Dysregulated hepatic bile acids collaboratively promote liver carcinogenesis** *Int J Cancer* **139**:1764–75 [Google Scholar](#)
36. Bindea G., et al. (2009) **ClueGO: a Cytoscape plug-in to decipher functionally grouped gene ontology and pathway annotation networks** *Bioinformatics* **25**:1091–3 [Google Scholar](#)
37. Koga H., et al. (2001) **Involvement of p21(WAF1/Cip1), p27(Kip1), and p18(INK4c) in troglitazone-induced cell-cycle arrest in human hepatoma cell lines** *Hepatology* **33**:1087–97 [Google Scholar](#)
38. Ma W.L., et al. (2014) **Androgen receptor roles in hepatocellular carcinoma, fatty liver, cirrhosis and hepatitis** *Endocr Relat Cancer* **21**:R165–82 [Google Scholar](#)
39. Schaefer K.L., et al. (2005) **Peroxisome proliferator-activated receptor gamma inhibition prevents adhesion to the extracellular matrix and induces anoikis in hepatocellular carcinoma cells** *Cancer Res* **65**:2251–9 [Google Scholar](#)
40. Zhang M., et al. (2015) **Nrf2 is a potential prognostic marker and promotes proliferation and invasion in human hepatocellular carcinoma** *BMC Cancer* **15**:531 [Google Scholar](#)
41. Dong T., et al. (2017) **FOXO1 inhibits the invasion and metastasis of hepatocellular carcinoma by reversing ZEB2-induced epithelial-mesenchymal transition** *Oncotarget* **8**:1703–1713 [Google Scholar](#)
42. Hashash J.G., Thudi K., Malik S.M. (2012) **An 18-year-old woman with a 15-cm liver mass and an ammonia level of 342** *Gastroenterology* **143**:1157 [Google Scholar](#)
43. Koo M., et al. (2017) **Biopsy-proven Hepatocellular Carcinoma in a 53-year-old Woman With Arginase Deficiency** *Pediatr Dev Pathol* **20**:517–521 [Google Scholar](#)
44. Rabinovich S., et al. (2015) **Diversion of aspartate in ASS1-deficient tumours fosters de novo pyrimidine synthesis** *Nature* **527**:379–383 [Google Scholar](#)
45. Wilson J.M., et al. (2012) **Hepatocellular carcinoma in a research subject with ornithine transcarbamylase deficiency** *Mol Genet Metab* **105**:263–5 [Google Scholar](#)
46. Lam T.L., et al. (2009) **Recombinant human arginase inhibits proliferation of human hepatocellular carcinoma by inducing cell cycle arrest** *Cancer Lett* **277**:91–100 [Google Scholar](#)
47. Wu T., et al. (2021) **Discovery of a Carbamoyl Phosphate Synthetase 1-Deficient HCC Subtype With Therapeutic Potential Through Integrative Genomic and Experimental Analysis** *Hepatology* **74**:3249–3268 [Google Scholar](#)

48. Della Torre S., et al. (2016) **An Essential Role for Liver ERalpha in Coupling Hepatic Metabolism to the Reproductive Cycle** *Cell Rep* **15**:360–71 [Google Scholar](#)
49. Bossard R., et al. (1993) **Ethinylestradiol treatment induces multiple canalicular membrane transport alterations in rat liver** *J Clin Invest* **91**:2714–20 [Google Scholar](#)
50. Stieger B., et al. (2000) **Drug- and estrogen-induced cholestasis through inhibition of the hepatocellular bile salt export pump (Bsep) of rat liver** *Gastroenterology* **118**:422–30 [Google Scholar](#)
51. Yamamoto Y., et al. (2006) **Estrogen receptor alpha mediates 17alpha-ethinylestradiol causing hepatotoxicity** *J Biol Chem* **281**:16625–31 [Google Scholar](#)
52. Hassan M.M., et al. (2017) **Estrogen Replacement Reduces Risk and Increases Survival Times of Women With Hepatocellular Carcinoma** *Clin Gastroenterol Hepatol* **15**:1791–1799 [Google Scholar](#)
53. Wang C.H., et al. (2022) **Hormone replacement therapy is associated with reduced hepatocellular carcinoma risk and improved survival in postmenopausal women with hepatitis B: A nationwide long-term population-based cohort study** *PLoS One* **17**:e0271790 [Google Scholar](#)
54. Simon F.R., et al. (2004) **Multihormonal regulation of hepatic sinusoidal Ntcp gene expression** *Am J Physiol Gastrointest Liver Physiol* **287**:G782–94 [Google Scholar](#)
55. De Witt E.H., Lack L. (1980) **Effects of sulfation patterns on intestinal transport of bile salt sulfate esters** *Am J Physiol* **238**:G34–9 [Google Scholar](#)
56. Alnouti Y., Klaassen C.D. (2011) **Mechanisms of gender-specific regulation of mouse sulfotransferases (Sults)** *Xenobiotica* **41**:187–97 [Google Scholar](#)
57. Heuman D.M (1989) **Quantitative estimation of the hydrophilic-hydrophobic balance of mixed bile salt solutions** *J Lipid Res* **30**:719–30 [Google Scholar](#)
58. Stiehl A., Earnest D.L., Admirant W.H. (1975) **Sulfation and renal excretion of bile salts in patients with cirrhosis of the liver** *Gastroenterology* **68**:534–44 [Google Scholar](#)
59. Cameron R.G., et al. (1982) **Promotive effects of steroids and bile acids on hepatocarcinogenesis initiated by diethylnitrosamine** *Cancer Res* **42**:2426–8 [Google Scholar](#)
60. Massafra V., et al. (2017) **Farnesoid X Receptor Activation Promotes Hepatic Amino Acid Catabolism and Ammonium Clearance in Mice** *Gastroenterology* **152**:1462–1476 [Google Scholar](#)
61. Schwarz M., et al. (2001) **Alternate pathways of bile acid synthesis in the cholesterol 7alpha-hydroxylase knockout mouse are not upregulated by either cholesterol or cholestyramine feeding** *J Lipid Res* **42**:1594–603 [Google Scholar](#)
62. Hoenerhoff M.J., et al. (2011) **Global gene profiling of spontaneous hepatocellular carcinoma in B6C3F1 mice: similarities in the molecular landscape with human liver cancer** *Toxicol Pathol* **39**:678–99 [Google Scholar](#)
63. Grinberg M., et al. (2014) **Toxicogenomics directory of chemically exposed human hepatocytes** *Arch Toxicol* **88**:2261–87 [Google Scholar](#)

64. Wang X., et al. (2019) **Identification of prognostic biomarkers for patients with hepatocellular carcinoma after hepatectomy** *Oncol Rep* **41**:1586–1602 [Google Scholar](#)
65. Cui H., et al. (2023) **Identification of a bile acid and bile salt metabolism-related lncRNA signature for predicting prognosis and treatment response in hepatocellular carcinoma** *Sci Rep* **13**:19512 [Google Scholar](#)
66. Huang H., et al. (2020) **Integrated analysis of microbiome and host transcriptome reveals correlations between gut microbiota and clinical outcomes in HBV-related hepatocellular carcinoma** *Genome Med* **12**:102 [Google Scholar](#)
67. Thomas C.E., et al. (2021) **Association between Pre-Diagnostic Serum Bile Acids and Hepatocellular Carcinoma: The Singapore Chinese Health Study** *Cancers* **13** [Google Scholar](#)
68. Degirolamo C., et al. (2015) **Prevention of spontaneous hepatocarcinogenesis in farnesoid X receptor-null mice by intestinal-specific farnesoid X receptor reactivation** *Hepatology* **61**:161–70 [Google Scholar](#)
69. Lee J.S., et al. (2004) **Classification and prediction of survival in hepatocellular carcinoma by gene expression profiling** *Hepatology* **40**:667–76 [Google Scholar](#)
70. Lee J.S., et al. (2004) **Application of comparative functional genomics to identify best-fit mouse models to study human cancer** *Nat Genet* **36**:1306–11 [Google Scholar](#)
71. Lee J.S., et al. (2006) **A novel prognostic subtype of human hepatocellular carcinoma derived from hepatic progenitor cells** *Nat Med* **12**:410–6 [Google Scholar](#)
72. Kim S.M., et al. (2012) **Sixty-five gene-based risk score classifier predicts overall survival in hepatocellular carcinoma** *Hepatology* **55**:1443–52 [Google Scholar](#)
73. Park Y.Y., et al. (2016) **Yes-associated protein 1 and transcriptional coactivator with PDZ-binding motif activate the mammalian target of rapamycin complex 1 pathway by regulating amino acid transporters in hepatocellular carcinoma** *Hepatology* **63**:159–72 [Google Scholar](#)
74. Roessler S., et al. (2010) **A unique metastasis gene signature enables prediction of tumor relapse in early-stage hepatocellular carcinoma patients** *Cancer Res* **70**:10202–12 [Google Scholar](#)
75. Sung W.K., et al. (2012) **Genome-wide survey of recurrent HBV integration in hepatocellular carcinoma** *Nat Genet* **44**:765–9 [Google Scholar](#)
76. Villa E., et al. (2016) **Neoangiogenesis-related genes are hallmarks of fast-growing hepatocellular carcinomas and worst survival. Results from a prospective study** *Gut* **65**:861–9 [Google Scholar](#)
77. Cancer Genome Atlas Research Network (2017) **Electronic address, W.B.E. and N. Cancer Genome Atlas Research, Comprehensive and Integrative Genomic Characterization of Hepatocellular Carcinoma.** *Cell* **169**:1327–1341 [Google Scholar](#)
78. Gertz J., et al. (2013) **Distinct properties of cell-type-specific and shared transcription factor binding sites** *Mol Cell* **52**:25–36 [Google Scholar](#)

79. Gordon F.K., et al. (2014) **Research resource: Aorta- and liver-specific ERalpha-binding patterns and gene regulation by estrogen** *Mol Endocrinol* **28**:1337–51 [Google Scholar](#)
80. Palierne G., et al. (2016) **Changes in Gene Expression and Estrogen Receptor Cistrome in Mouse Liver Upon Acute E2 Treatment** *Mol Endocrinol* **30**:709–32 [Google Scholar](#)

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Reviewer #1 (Public review):

Liver cancer shows a high incidence in males than females with incompletely understood causes. This study utilized a mouse model that lacks the bile acid feedback mechanisms (FXR/SHP DKO mice) to study how dysregulation of bile acid homeostasis and a high circulating bile acid may underlie the gender-dependent prevalence and prognosis of HCC. By transcriptomics analysis comparing male and female mice, unique sets of gene signatures were identified and correlated with HCC outcomes in human patients. The study showed that ovariectomy procedure increased HCC incidence in female FXR/SHP DKO mice that were otherwise resistant to age-dependent HCC development, and that removing bile acids by blocking intestine bile acid absorption reduced HCC progression in FXR/SHP DKO mice. Based on these findings, the authors suggest that gender-dependent bile acid metabolism may play a

role in the male-dominant HCC incidence, and that reducing bile acid level and signaling may be beneficial in HCC treatment. This study include many strengths: 1. Chronic liver diseases often proceed the development of liver and bile duct cancer. Advanced chronic liver diseases are often associated with dysregulation of bile acid homeostasis and cholestasis. This study takes advantage of a unique FXR/SHP DKO model that develop high organ bile acid exposure and spontaneous age-dependent HCC development in males but not females to identify unique HCC-associated gene signatures. The study showed that the unique gene signature in female DKO mice that had lower HCC incidence also correlated with lower grade HCC and better survival in human HCC patients. 2. The study also suggests that differentially regulated bile acid signaling or gender-dependent response to altered bile acids may contribute to gender-dependent susceptibility to HCC development and/or progression. 3. The sex-dependent differences in bile acid-mediated pathology clearly exist but are still not fully understood at the mechanistic level. Female mice have been shown to be more sensitive to bile acid toxicity in a few cholestasis models, while this study showed a male dominance of bile acid promotion of HCC. This study used ovariectomy to demonstrate that female hormones are possible underlying factors. Future studies are needed to understand the interaction of sex hormones, bile acids, and chronic liver diseases and cancer.

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Author response:

The following is the authors' response to the previous reviews

Reviewer #1 (Public review):

Summary:

Liver cancer shows a high incidence in males than females with incompletely understood causes. This study utilized a mouse model that lacks the bile acid feedback mechanisms (FXR/SHP DKO mice) to study how dysregulation of bile acid homeostasis and a high circulating bile acid may underlie the gender-dependent prevalence and prognosis of HCC. By transcriptomics analysis comparing male and female mice, unique sets of gene signatures were identified and correlated with HCC outcomes in human patients. The study showed that ovariectomy procedure increased HCC incidence in female FXR/SHP DKO mice that were otherwise resistant to agedependent HCC development, and that removing bile acids by blocking intestine bile acid absorption reduced HCC progression in FXR/SHP DKO mice. Based on these findings, the authors suggest that gender-dependent bile acid metabolism may play a role in the male-dominant HCC incidence, and that reducing bile acid level and signaling may be beneficial in HCC treatment.

strengths:

(1) Chronic liver diseases often proceed the development of liver and bile duct cancer. Advanced chronic liver diseases are often associated with dysregulation of bile acid homeostasis and cholestasis. This study takes advantage of a unique FXR/SHP DKO model that develop high organ bile acid exposure and spontaneous age-dependent HCC development in males but not females to identify unique HCC-associated gene signatures. The study showed that the unique gene signature in female DKO mice that had lower HCC incidence also correlated with lower grade HCC and better survival in human HCC patients. 2. The study also suggests that differentially regulated bile acid signaling or gender-dependent response to altered bile acids may contribute to gender-dependent susceptibility to HCC development and/or progression. 3. The sex-dependent differences in bile acidmediated pathology clearly exist but are still not fully understood at the mechanistic level. Female mice have been shown to be more sensitive to bile acid

toxicity in a few cholestasis models, while this study showed a male dominance of bile acid promotion of HCC. This study used ovariectomy to demonstrate that female hormones are possible underlying factors. Future studies are needed to understand the interaction of sex hormones, bile acids, and chronic liver diseases and cancer.

We thank Reviewer 1 for their positive and thorough assessment of our manuscript

Weaknesses:

(1) HCC shows heterogeneity, and it is unclear what tissues (tumor or normal) were used from the DKO mice and human HCC gene expression dataset to obtain the gene signature, and how the authors reconcile these gene signatures with HCC prognosis.

Mice studies: Aged DKO mice develop aggressive tumors (major and minor nodules, See Figure 1), and the entire liver is burdened with multiple tumor nodules. It is technically challenging to demarcate the tumor boundaries as most of the surrounding tissues do not display normal tissue architecture. Therefore, livers from age- and sexmatched wild-type C57/BL6 mice were used as control tissue. All the mice were inbred in our facility. Spatial transcriptomics and longitudinal studies are ongoing to collect tumors at earlier time points wherein we can differentiate tumor and non-tumor tissue.

Human Studies: We mined five separate clinical data sets. The human HCC gene expression comprised of samples from the (i) National Cancer Institute (NCI) cohort (GEO accession numbers, GSE1898 and GSE4024) and (ii) Korea, (iii) Samsung, (iv) Modena, and (v) Fudan cohorts as previously described (GEO accession numbers, GSE14520, GSE16757, GSE43619, GSE36376, and GSE54236). We have added a new supplemental table 4, giving details of these datasets. Depending on the cohort, they are primarily HCC samples- surgical resections of HCC, control samples, with some tumors and paired non-tumor tissues.

(2) The authors identified a unique set of gene expression signatures that are linked to HCC patient outcomes, but analysis of these gene sets to understand the causes of cancer promotion is still lacking. The studies of urea cycle metabolism and estrogen signaling were preliminary and inconclusive. These mechanistic aspects may be followed up in revision or future studies.

We agree. Experiments to elicit HCC causality and promotion are complex, given the heterogeneous nature of liver cancer. Moreover, the length of time (12 months) needed to spontaneously develop cancer in this DKO mouse model makes it challenging. As mentioned by the reviewer, mechanistic studies are ongoing, and longitudinal time course experiments are actively being pursued to delineate causality. Having said that, we mined the TCGA LIHC (The Cancer Genome Atlas Liver Hepatocellular Carcinoma) database to examine the expression of the individual urea cycle genes and found them suppressed in liver tumorigenesis (new Supplementary Figure 4). We also evaluated if estrogen receptor α (Er α) targets altered in DKO females (DKO_Estrogen) correlate with overall survival in HCC (new Supplementary Figure 6). We note that Er α expression per se is reduced in males and females upon liver tumorigenesis. Also, DKO_Estrogen signature positively corroborated with better overall survival (new Supplementary Figure 6). These findings further bolster the relevance of urea cycle metabolism and estrogen signaling during HCC.

(3) While high levels of bile acids are convincingly shown to promote HCC progression, their role in HCC initiation is not established. The DKO model may be limited to conditions of extremely high levels of organ bile acid exposure. The DKO mice do not model the human population of HCC patients with various etiology and shared liver pathology (i.e. cirrhosis). Therefore, high circulating bile acids may not fully explain the male prevalence of HCC incidence.

We agree with this comment that our studies do not show bile acids can initiate HCC and may act as one of the many factors that contribute to the high male prevalence of HCC. This is exactly the reason why throughout the manuscript we do not write about HCC initiation. To clarify further, in the revised discussion of the manuscript, we have added a sentence to highlight this aspect, “while this study demonstrates bile acids promote HCC progression it does not investigate or provide evidence if excess bile acids are sufficient for HCC initiation.”

(4) The authors showed lower circulating bile acids and increased fecal bile acid excretion in female mice and hypothesized that this may be a mechanism underlying the lower bile acid exposure that contributed to lower HCC incidence in female DKO mice. Additional analysis of organ bile acids within the enterohepatic circulation may be performed because a more accurate interpretation of the circulating bile acids and fecal bile acids can be made in reference to organ bile acids and total bile acid pool changes in these mice.

As shown in this manuscript- we provide BA compositional analyses from the liver, serum, urine, and feces (Figures 5 and 6, new Supplementary Figure 8, Supplementary Tables 4 and 5). Unfortunately, we did not collect the intestinal tissue or gallbladders for BA analysis in this study. Separate cohorts of mice are being aged for future BA analyses from different organs within the enterohepatic loop. We thank you for this suggestion. Nevertheless, we have previously measured and reported BA values to be elevated in the intestines and the gall bladder of young DKO mice (PMC3007143).

Reviewer #2 (Public review):

Weaknesses:

(1) The translational value to human HCC is not so strong yet. Authors show that there is a correlation between the female-selective gene signature and low-grade tumors and better survival in HCC patients overall. However, these data do not show whether this signature is more highly correlated with female tumor burden and survival. In other words, whether the mechanisms of female protection may be similar between humans and mice. In that respect, it would also be good to elaborate on whether women have higher fecal BA excretion and lower serum BA concentration.

The reviewer poses an interesting question to test if the DKO female-specific signatures are altered differently in male vs. female HCC samples. As we found the urea cycle and estrogen signaling to be protective and enriched in our mouse model, we tested their expression pattern using the TCGA-LIHC RNA-seq data. We found urea cycle genes and *Er* transcripts broadly reduced in tumor samples irrespective of the sex (new Supplementary Figure 4 and Supplementary Figure 6), indicating that these pathways are compromised upon tumorigenesis even in the female livers.

While prior studies have shown (i) a smaller BA pool w synthesis in men than women (PMID: 22003820), we did not find a study that systematically investigated BA excretion between the sexes in HCC context. The reviewer is spot on in suggesting BA analysis from HCC and unaffected human fecal samples from both sexes. Designing and performing such studies in the future will provide concrete proof of whether BA excretion protects female livers from developing liver cancer. We thank you for these suggestions.

(2) The authors should perform a thorough spelling and grammar check.

We apologize for the typos, which have been fixed, and as suggested by the reviewer, we have performed a grammar check.

(3) There are quite some errors and inaccuracies in the result section, figures, and legends. The authors should correct this.

We apologize for the inadvertent errors in the manuscript, and we have clarified these inaccuracies in the revised version. Thank you.

Reviewer#1 (Recommendations for the authors).

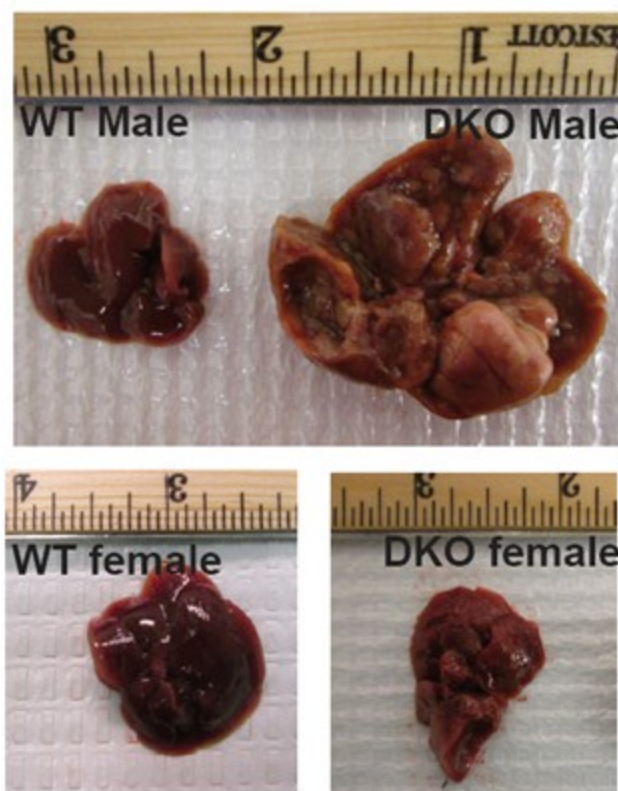
(1) Figures 1A-F, This statement of altered liver steatosis needs to be further supported by measurement of liver triglycerides. Lower magnification images of Sirius red stain should be shown for better evaluation of liver fibrosis.

Unfortunately, we did not measure liver triglycerides and sirius red stained samples have faded, and lower magnification is unavailable at this juncture. We have modified our results accordingly.

We did not take the gross picture of WT female and DKO female livers in the same frame as shown below. Since the manuscript is focused on male and female differences in liver cancer incidence, we provided DKO male and female liver images as Figure 1D in the paper.

Author response image 1.

Gross liver images of a year-old WT and DKO mice which show prominent hepatocarcinogenesis in DKO male mice



(2) Can the authors clarify if the gene transcriptomics was performed with normal or tumor tissues of DKO mice?

Gene transcriptomics were performed with the tumor tissue of DKO mice. We have previously published data from younger non tumor bearing DKO male mice (PMCID: PMC3007143).

(3) Supplementary Figure 3C. Could the authors confirm if this is F vs M or just DKO female since it does not seem to match the result description in the main text? It is better practice to indicate the sub-panels of the Supplementary Figures in the main text while describing the results.

As the reviewer correctly points out Supplementary Figure 3C is DKO F vs M signature not DKO_female signature and this has been clarified in the text. We have also included DKO_F data now to reduce the confusion.

(4) Figure 3. Legend, the data presented are not well explained in the Legend, especially the labeling and what is being presented and compared.

As suggested by the reviewer, we have modified the legend accordingly.

(5) Supplementary Table 4 does not contain total serum bile acid as described in the main text.

We agree with the reviewer. We provided primary and secondary BA concentrations, Supplementary Table 4 (currently Supplementary Table 5 in the revised version): Rows 20 and 21. but not their added total. We have modified the text accordingly.

(6) Method section: many experiments lack descriptions of details.

We have added details to the animal experimental design, ER CHIP-PCR, schematics of experiments are included within the main and supplemental figures, metabolomics and BA analysis have been expanded.

Reviewer #2 (Recommendations For The Authors):

General:

(1) The authors are advised to do a thorough grammar and spelling check.

We have performed spelling and grammar check as suggested using an online platform Grammarly. Thank You.

Results:

(1) Figure 1 o The authors should show in Figure 1D female WT and female DKO liver.

See Figure 1 added in our responses to point 1 of reviewer 1's comment.

In the Figure legend, (A-E) should be replaced by (A+D).

Thank you. We have modified it accordingly.

The authors do not refer to 1J in the text, please add this reference.

Thank you for pointing it. We have referenced 1J in the text.

The description of 1H does not elaborate on the sex differences in ALT/AST levels, as this is the focus of the manuscript.

We have added a sentence to show that the injury markers are higher in DKO males, which is consistent with an advanced disease. Thanks.

The authors should use the correct nomenclature in Figure 1I/1J (gene vs protein and capitals vs non-capitals).

The Figure 1I and 1J show gene expression of Fxr and Shp and hence we used the non-capital italicized nomenclature. Thanks.

(2) Figure 2:

The x-axis length is different in Figures 2A and 2B. Please correct to visualize the differences between males and females better.

The x axis length has been fixed as suggested. Thanks

(3) Figure 3:

The authors should elaborate on how the patients were assigned to each gene signature. This is not fully clear.

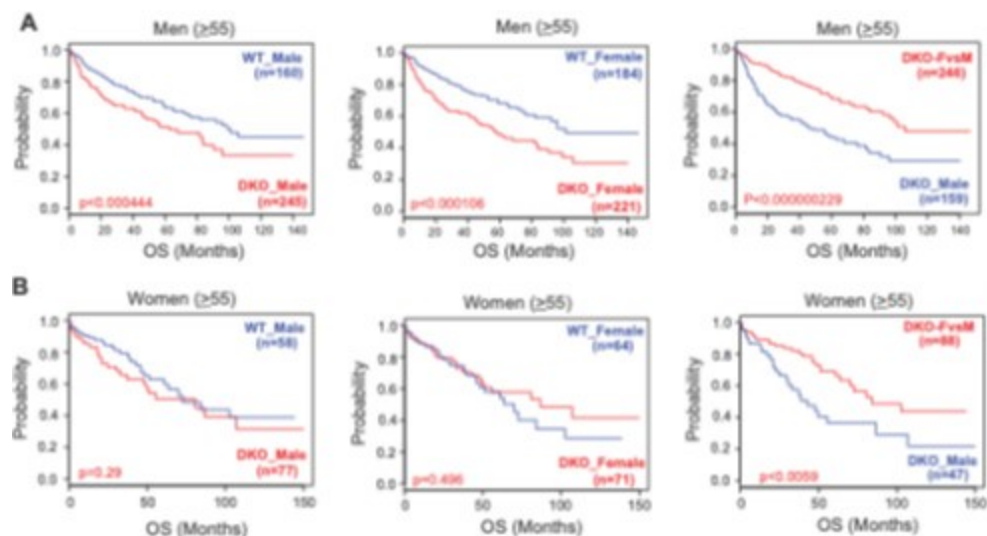
The gene set obtained from the WT and DKO mice were used. The process used is shown as a schematic in Supplemental Fig 2C and the gene list is included in an excel sheet as Supplemental table 1.

We are curious how these data (F3A-C) would look when separating male and female human patients.

We performed an overall survival analysis with a subgroup of patients and provide it. We segregated the HCC cohort data on sex and age (>55 yr, since we assumed 55 as an age for menopause) and evaluated the DKO gene signature. Similar to the original figure 3, we find that irrespective of sex, and age, DKO FvsM gene signature corresponds with better overall survival in men and in women. These findings align with the combined analysis in overall survival shown in original Figure 3 of the manuscript, and therefore we did not modify it. If deemed necessary, we are happy to include the figure below to reviewers in the main manuscript.

Author response image 2.

Correlation of gene signatures obtained from WT and DKO mouse model with the survival data of HCC patients segregated by age and sex. The Kaplan Meier Survival graphs were generated based on WT and DKO transcriptome changes using five HCC clinical cohorts. Analysis of OS (Overall Survival) in patients ((A) Men and (B) Women) using the gene signatures representative of either male WT or male DKO, female WT or female DKO, and unique changes observed in female DKO mice but not in male DKO mice.



What was used as the control signature in Figure 3C? Please specify this.

For Figure 3C we compared the DKO_M signature to that of DKOF vs M signature. These genes are listed as an Excel Sheet (Supplementary Table 1).

The authors claim that DKO female mice display chronic cholestasis, similar to their male counterparts. Please refer to previous work or show the data.

Serum BA levels are elevated in DKO females are reported in supplementary table 5 and we find comparable hepatic BA composition in Figure 5 F.

(4) Figure 4: Labels for the x-axis are missing in Figure 4C. Please add legends or labels to the bars.

The x axis label is included in the top Serum BAs in (□M)

In Figure 4I, the percentage of input is quite low. An IgG control would show whether recruitment of ERalpha to the shown loci is significant above background levels. Also, ChIP on the OVX liver could serve as a negative control.

We did use IgG as control pull down and the signals above this background were considered. We have not performed this in OVX, which would be an excellent negative control for future studies. Thank You.

The results and legends refer to ChIP-qPCR, while methods only mention ChIP-seq. Please adapt.

We sincerely apologize for the mistake. We used published ChIP-seq to identify putative binding site and then performed ChIP PCR to validate it. We have clarified and rectified this error. Thank You.

Significance indications in the figure legend do not correspond with significance indications in the figure. Please explain the used significance symbols in the figure in the legend.

Thank You. The legends and their significance have been matched.

(5) Figure 5:

Authors claim lowered total serum BA in females compared to males, and reference to Supplementary Table 4. However, these data are not provided, only percentages and ratios are displayed.

In the revised version, this has become Table 5. See response to the same concern noted by Reviewer 1, Point 5 above.

Figure 5D: Are sulphated BA also elevated in WT females? Please provide these data.

There is no significant urinary excretion of BAs in WT control animals. We have previously measured and found none. But under cholestatic conditions BAs are observed in urine. Therefore, sulphated BA levels were found only in the DKO mice.

Figure 5H: Is the fecal BA excretion in WT females also proportionally higher than in males? Please provide these data.

We were unable to perform the untargeted metabolomics profiling of WT fecal samples. When we measured for BAs in the feces, as expected very low conc were present irrespective of the sex ($\sim 0.01 \mu\text{M}$) and we did not find any sex difference. Also, prior studies in 129SVJ strain exhibited comparable fecal excretion (PMC150802). We did not find any clinical studies that measured fecal BA between the sexes.

(6) Figure 6:

References in the text of the result section to Figure 6 are wrong. The authors should change this.

Thank You. This has been rectified.

Significance indications in the legend do not correspond with significance indications in the figure. Please explain the used significance symbols in the figure in the legend.

Thank You. The legends and their significance have been matched.

(7) Supplemental Figure 3:

Please adapt the title of this figure; the sentence is incorrect. The description of this figure is very poor.

We have modified the legend and the title of the Supplemental Figure 3 to make it more appropriate. Thanks

Please explain what the blue and red dots represent.

Each dot in blue and yellow indicate the Bayesian probability generated from our BCCP model.

What are the bold horizontal lines representing? Why are there no dots in some box plots? Please elaborate.

The box represents the interquartile range (IQR), encompassing the middle 50% of the data. The bottom and top edges correspond to the 25th and 75th percentiles, respectively, while the

bold horizontal line indicates the median value.

The absence of visible dots in certain categories—particularly in higher CLIP and TNM stages—is due to the small number of patients, all of whom had similar Bayesian prediction probabilities. As these values cluster tightly around the median, the individual dots may be overlapped and hidden behind the median line.

| *The figure is not visually easy to understand, please reconsider the representation.*

We hope the modified figure legends with the explanation of the lines and the points in the graphs increases the clarity and makes them acceptable.

| *Please add the DKO_female signature plot.*

We have added these graph to Supplemental figure 3

| (8) Supplemental 4A:

| *Fold change at Z-score is missing. This should be added.*

Thank you we have added this information

| (9) Supplemental 5:

| *The scale bar is missing. This should be included.*

The figure is now supplemental figure 8 and the scale bar has been added.

| *Methods:*

| (1) *Did the authors use ChIP-sequencing or ChIP-qPCR? Please describe the correct method.*

We apologize for the error. We have used ChIP-PCR and rectified it in our methods and in our response to a figure 4 query.

| (2) *It is unclear how the mouse model was generated. Please refer to earlier publications.*

The mice were generated in house at UIUC, and we have added this sentence to the Methods section. The original reference has been cited in the text (PMCID: PMC3007143).

| *Discussion:*

| (1) *The authors claim in the discussion: 'consistently higher recruitment of ER to the classical BA synthetic genes ...' This is not shown in Figure 4I, only ER recruitment to Cyp7a1 is significantly higher in females. Please rephrase.*

We agree and we have modified the sentence Cyp7A1 accounts for ~75% of BA synthesis and is a rate-limiting gene in the classical BA synthesis pathway.

| (2) *The authors could make their statements stronger if they could elaborate on whether women have more fecal BA excretion, and if there are differences in serum BA concentration in HCC between male and female patients.*

Unfortunately, we were unable to find clinical studies with appropriate controls which examined and reported serum BA in HCC in a sex specific manner.

In addition, to understand whether the female-specific protections in humans are similar to mice, it would be nice to show correlations of the female-specific mouse signature with male and female liver signatures.

At this time, we do not have large n numbers of control or precancerous early-stage patient datasets from both sexes to make such comparisons. Nevertheless, there is translational relevance of these sex-specific signature. Figure 2 included in the reviewer response shows that DKO male signature correlates with poor overall survival in males, whereas neither DKO male nor DKO female signature predict outcome in females. In contrast, DKO female-specific gene signature (DKOFvsM) correlates with better overall survival in both men and in women.

(3) The authors state in the discussion: 'Currently we do not know how to reconcile this data other than indicating a potential ER independent mechanism.' We do not understand the reasoning behind this statement. Please clarify.

We find that increased Era expression in DKO coincides with CA-mediated suppression of BA synthesis genes in the absence of Fxr and Shp. But we also noticed that in OVX DKO mice, Era expression is blunted, and so is basal BA synthesis gene expression. Putting together these data, it is intriguing that Era expression correlates both positively and negatively with BA synthesis genes. To reconcile these contrasting results, we have written the following sentence in the discussion.

“These findings suggest Era expression is linked to both positive and negative regulation of BA synthesis genes. But we do not know how ER elicits these differential effects on BA synthesis.”

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